



Data Warehousing and Data Mining

Block-1 Unit-1

Fundamentals of Data Warehouse

Topics to be Covered

- Introduction
- Evolution of Data Warehouse
- Data Warehouse and its Need
- Data Warehouse Design Approaches
- Characteristics of a Data Warehouse
- OLTP and OLAP
- Data Granularity

Topics to be Covered

- Metadata and Data Warehousing
- Data Warehouse Applications
- Types of Data warehouse
- Popular Data Warehouses Platforms
- Important Topics
- Summary



<https://www.javatpoint.com/data-mining-cluster-vs-data-warehousing>

Introduction

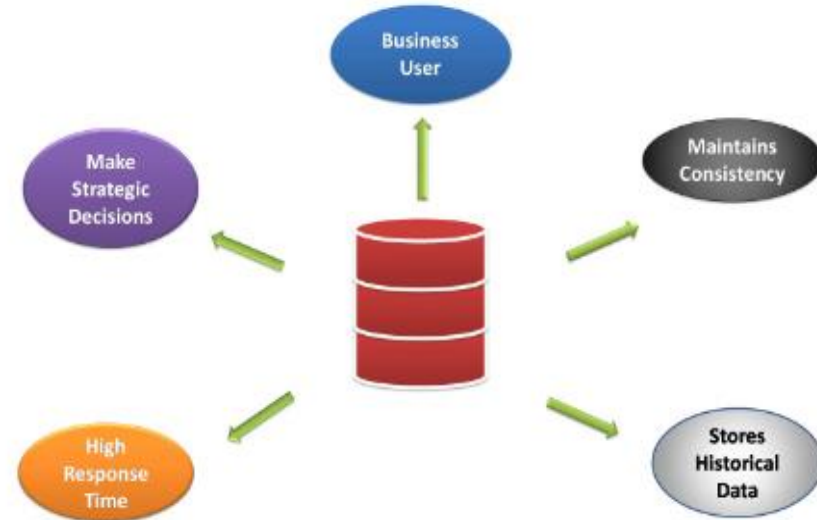
- A data warehouse is a central repository for storing large amounts of data.
- It is designed for business intelligence activities such as analysis and reporting. Key components include data sources, integration, storage, and access.
- Understanding these fundamentals is crucial for effective design, implementation, and usage of a data warehouse.

Evolution of Data Warehouse

- The relational database revolution in the early 1980s ushered in an era of improved access to the valuable information contained deep within data.
- It was soon discovered that databases modeled to be efficient at transactional processing were not always optimized for complex reporting or analytical needs.

Data Warehousing and its Need

- Data warehouse systems are real-time repositories of information, which are likely to be tied to specific applications.
- Data warehouses gather data from multiple sources with an emphasis on storing, filtering, retrieving and analyzing huge quantities of organized data.



https://www.google.com/search?rlz=1C1CHBD_enIN1056IN1056&q=Data+Warehousing+%26+Its+Need&tbm=isch&sa=X&ved=2ahUKewi0o4XOgsX_AhV-

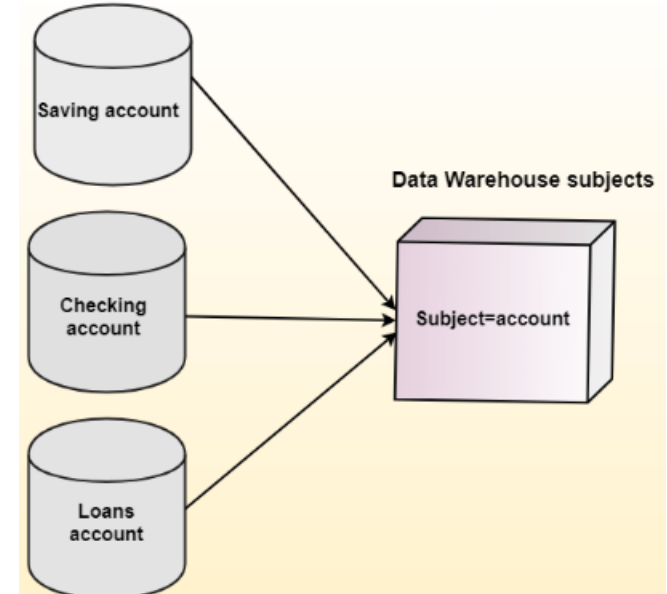
Data Warehousing and its Need

Need for Data Warehouse:

- Enhancing the turnaround time for analysis and reporting
- Improved Business Intelligence
- Benefit of historical data
- Standardization of data
- Immense ROI

Data Warehousing and its Need

- **Benefits of Data Warehouse :**
 - Scalability
 - Access to Historical Insights
 - Works On-Premises and on Cloud
 - Better Efficiency
 - Improved Data Security
 - Increase Revenue and Returns



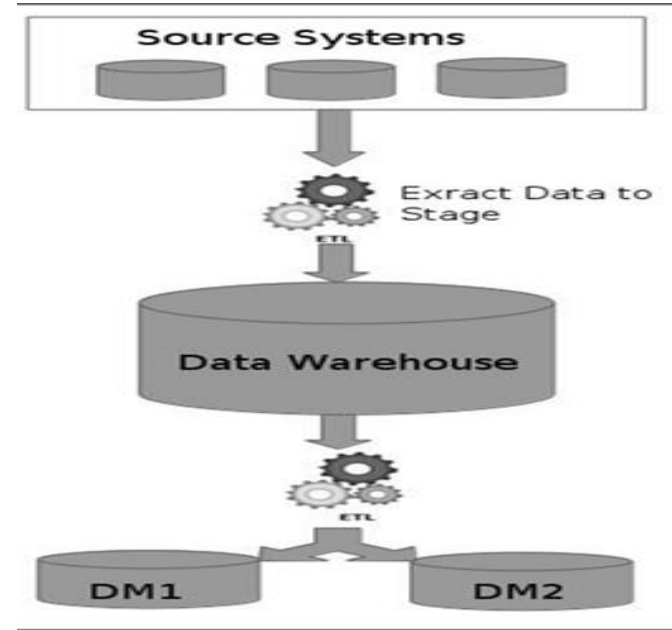
<https://www.javatpoint.com/data-warehouse>

Data Warehouse Design Approaches

- Data Warehouse design approaches are very important aspect of building data warehouse. Selection of right data warehouse design could save lot of time and project cost. There are two different Data Warehouse Design Approaches
 - Top-down Approach
 - Bottom-up Approach

Data Warehouse Design Approaches

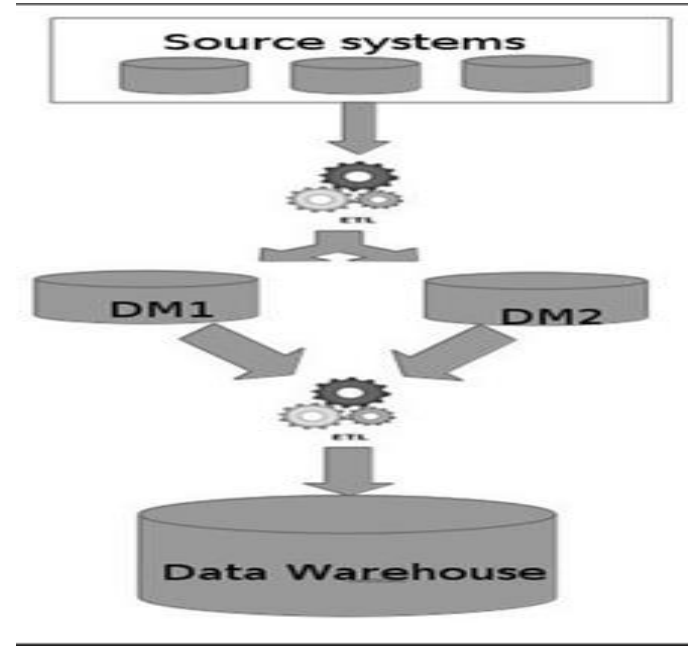
- **Top-down Approach**
 - In the top-down approach, the data warehouse is designed first and then data mart is built on top of data warehouse.



Data Warehouse Design Approaches

- **Bottom-up Approach**

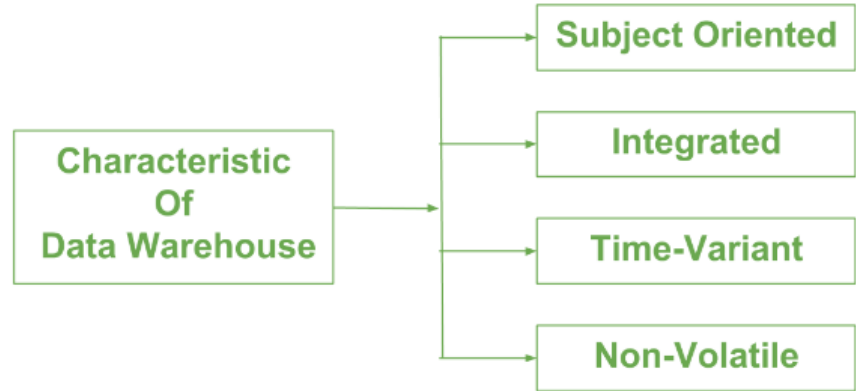
- In the Bottom-up Approach , data marts are first created to provide the reporting and analytics capability for specific business process, later with these data marts enterprise data warehouse is created.



Characteristics of a Data Warehouse

- The data warehouses have four essential characteristics that distinguish them from any other data and these characteristics are as follows:

- Subject-oriented
- Integrated
- Time-Variant
- Non-Volatile

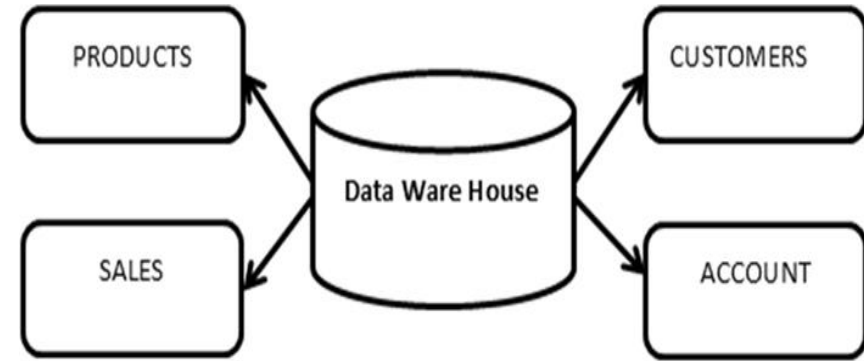


<https://www.geeksforgeeks.org/characteristics-and-functions-of-data-warehouse/>

Characteristics of a Data Warehouse

- **Subject-Oriented:**

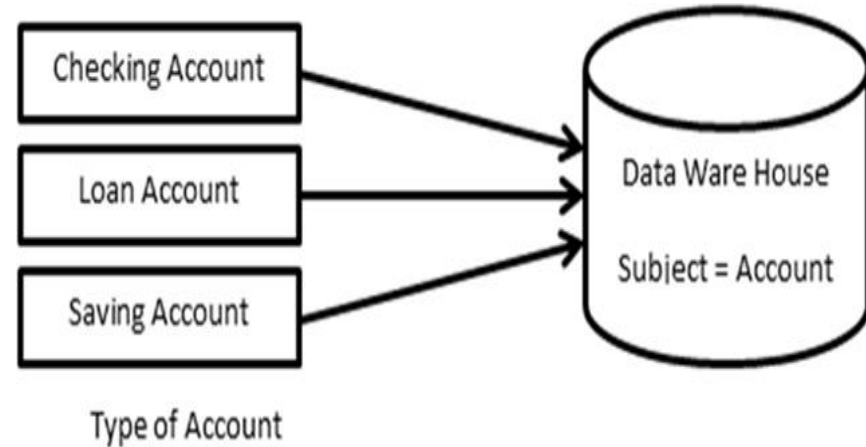
- A DW is always a subject-oriented one, as it provides information about a specific theme instead of current organizational operations .
- It also provides an easy and accurate demonstration of specific themes by eliminating information that is not needed to make decisions.



Characteristics of a Data Warehouse

- **Integrated:**

- Integration involves setting up a common system to measure all similar data from multiple systems.

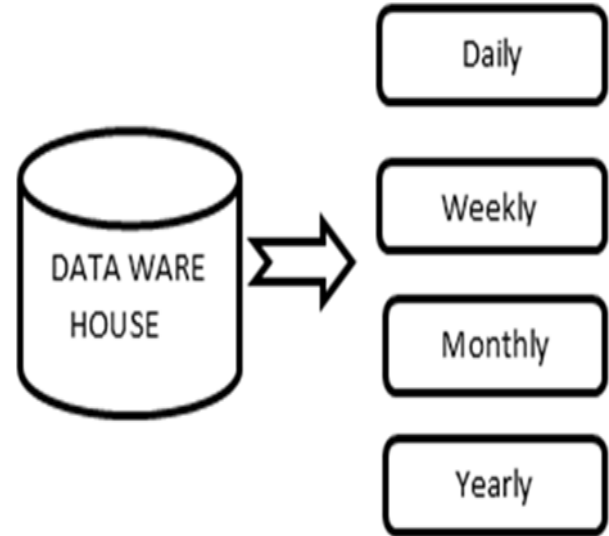


- Data was to be shared within several database repositories and must be stored in a secured manner to access by the data warehouse.

Characteristics of a Data Warehouse

- **Time-Variant:**

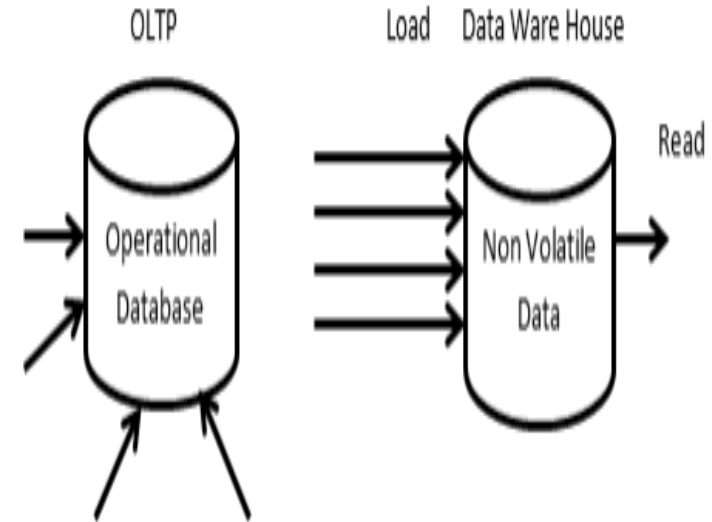
- In the time variant Information may be held in various intervals such as weekly, monthly, and yearly and It provides a series of limited-time, variable rate, online transactions.



Characteristics of a Data Warehouse

- **Non-Volatile:**

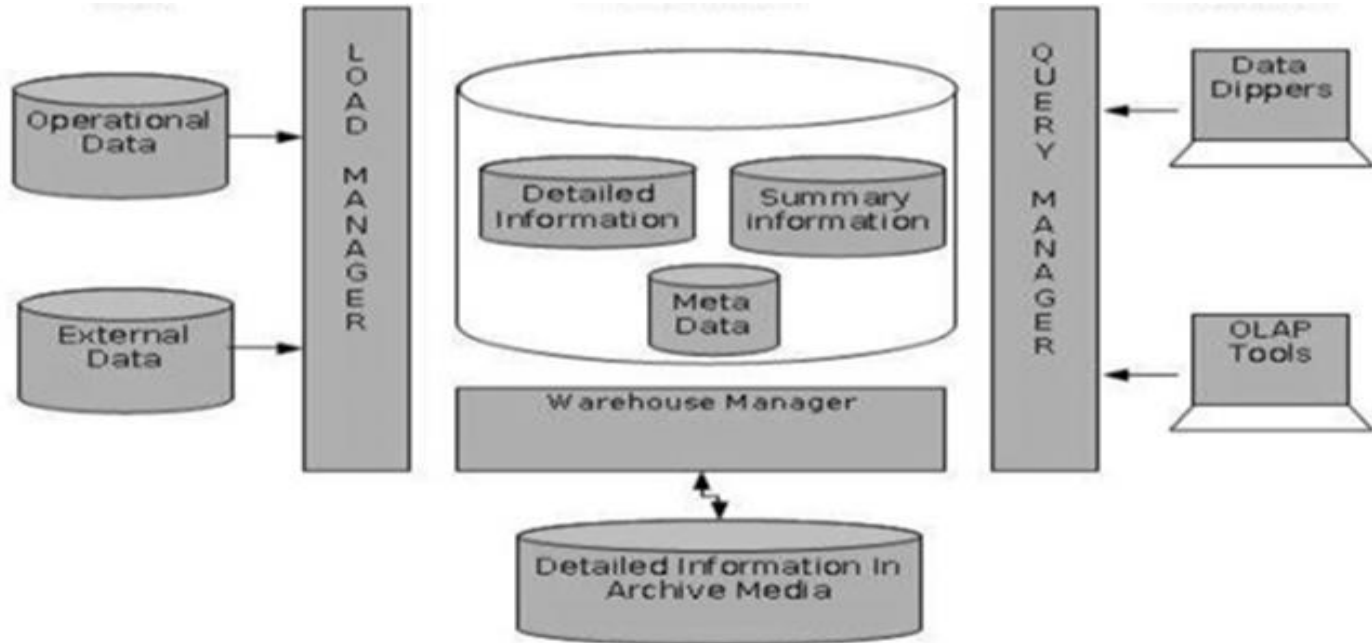
- The data residing in the data warehouse is permanent, as the name non - volatile suggests. It also ensures that when new data is added, data is not erased or removed.



How Data Warehouse Works?

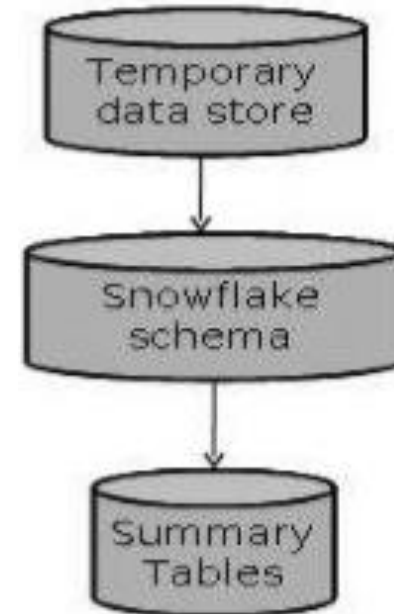
- A data warehouse is a central repository in which one or more sources of information are collected.
- Data in the data warehouse may be Structured, Semi-structured, or Unstructured.
- Data are processed, transformed, and accessed by end users for use in business intelligence reporting and decision-making.

How Data Warehouse Works?



How Data Warehouse Works?

- **Important components of the data warehouse:**
 - Load Manager
 - Warehouse Manager
 - Query Manager



How Data Warehouse Works?

- **Load Manager:**

- Load Manager Component of data warehouse is responsible for collection of data from operational system and converts them into usable form for the users.
- This component is responsible for importing and exporting data from operational systems.
- This component includes all of the programs and applications interfaces that are responsible for pooling the data out of the operational system.

How Data Warehouse Works?

- **Warehouse Manager:**
 - The warehouse manager is the center of data-warehousing system and is the data warehouse itself. It is a large, physical database that holds a vast amount of information from a wide variety of sources.
 - The data within the data warehouse is organized such that it becomes easy to find, use and update frequently from its sources.

How Data Warehouse Works?

- **Query Manager :**
 - Query Manager Component provides the end-users with access to the stored warehouse information through the use of specialized end-user tools.
 - Data mining access tools have various categories such as query and reporting, online analytical processing (OLAP), statistics, data discovery and graphical and geographical information systems.

How Data Warehouse Works?

- **End-user access tools** : This is divided into the following categories, such as :-
 - Reporting Data
 - Query Tools
 - Data Dippers
 - Tools for EIS
 - Tools for OLAP and tools for data mining

OLTP and OLAP

- Online Transaction Processing (OLTP) and Online Analytical Processing (OLAP) are the two terms which look similar but refer to different kinds of system components of the data warehouse:-
- **Online Transaction Processing (OLTP) :-**
 - An OLTP system captures and maintains transaction data in a database. Each transaction involves individual database records made up of multiple fields or columns.

OLTP and OLAP

- **Online Analytical Processing (OLAP):**
 - OLAP applies complex queries to large amounts of historical data, aggregated from OLTP databases and other sources, for data mining, analytics, and business intelligence projects.
 - Each query involves one or more columns of data aggregated from many rows.

Data Granularity

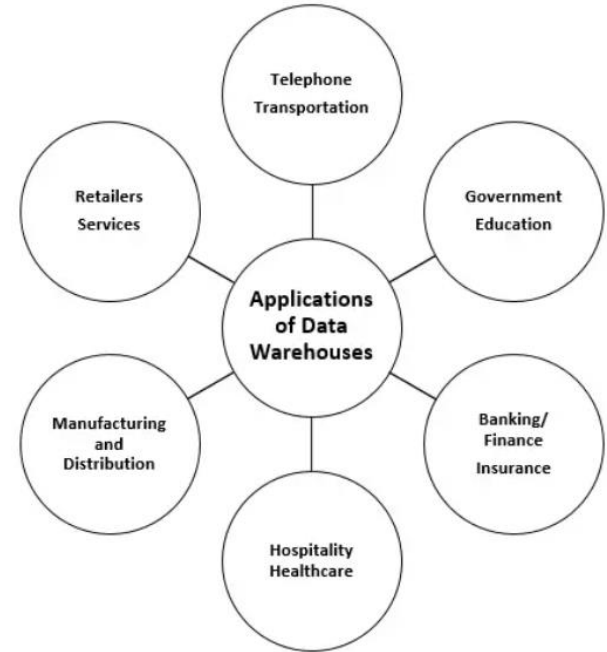
- Granularity is one of the main elements in the modeling of DW data. Granularity of data refers to detail levels.
- Multiple levels of detail may be available depending on the requirements. At least two granular levels exist for many data warehouses.
- It means greater detail of the data (less summary).

Metadata & Data Warehousing

- In a data warehouse, data is stored using a common schema controlled by a common dictionary. Within the data dictionary, data is kept about the logical data structures, file and address data, index information and others. The metadata should contain the following data warehouse information:-
 - The data structure based on the programmer's view
 - Data structure based on DSS analysts' view
 - The DW's data sources

Data Warehousing Applications

- Some of the applications of data warehouses:-
 - Investment and Insurance
 - Healthcare
 - Retail
 - Social Media Websites
 - Banking
 - Government



<https://whatisdbs.com/12-applications-of-data-warehouse/>

Types of Data Warehouses

- There are three different types of traditional Data Warehouse models as listed below:
 - Enterprise
 - Operational
 - Data Mart



Types of Data Warehouses

- **Enterprise Data Warehouse:**

- An enterprise provides a central repository database for decision support throughout the enterprise.
- It is a central place where all business information from different sources and applications are made available .
- Once it is stored, it can be used for analysis and used by all the people across the organization.

Types of Data Warehouses

- **Operational Data Warehouse :**
 - These features have a sizable enterprise-wide scope, but unlike the substantial enterprise warehouse, data is refreshed in near real-time and used for routine commercial activity.
 - It assists in obtaining data straight from the database, which also helps data transaction processing.

Types of Data Warehouses

- **Data Mart**

- Data Mart may be a subset of knowledge warehouse, and it supports a specific region, business unit, or business function. Data Mart focuses on storing data for a particular functional area, and it contains a subset of data saved in a memory.

Popular Data Warehouse Platforms

Microsoft Azure Synapse

- Google Big Query

AWS Redshift

- Snowflake



<https://cxl.com/blog/bigquery/>

Important Topics

Characteristics of a data warehouse .

Difference between OLAP And OLTP

Data Warehouse Design Approaches.

Data Granularity

Summary

- **In this session we studied about -**
- Data Warehouse as a concept designed to aid strategic information.
- Data Warehouse allows people to make decisions and provides flexible, convenient and interactive sources of strategic intelligence.
- Data warehouses are meant to be used by executives, managers, and other people at higher managerial levels who may not have much technical expertise in handling the databases.

*Thank
You*



Data Warehousing and Data Mining

Block-1 Unit-3

Dimensional Modeling

Topics to be Covered

Introduction



Dimensional Modeling



Strengths of Dimensional Modeling



Identifying Facts and Dimensions



Star Schema



Advantages and Disadvantages of Star Schema



Snowflake Schema

Topics to be Covered

Advantages and Disadvantages of Snowflake Schema



Fact Constellation Schema



Aggregate Tables



Need for Building Aggregate Fact Tables



Limitations of Aggregate Fact Tables



Aggregate Fact Tables and Derived Dimension Tables



Important Topics



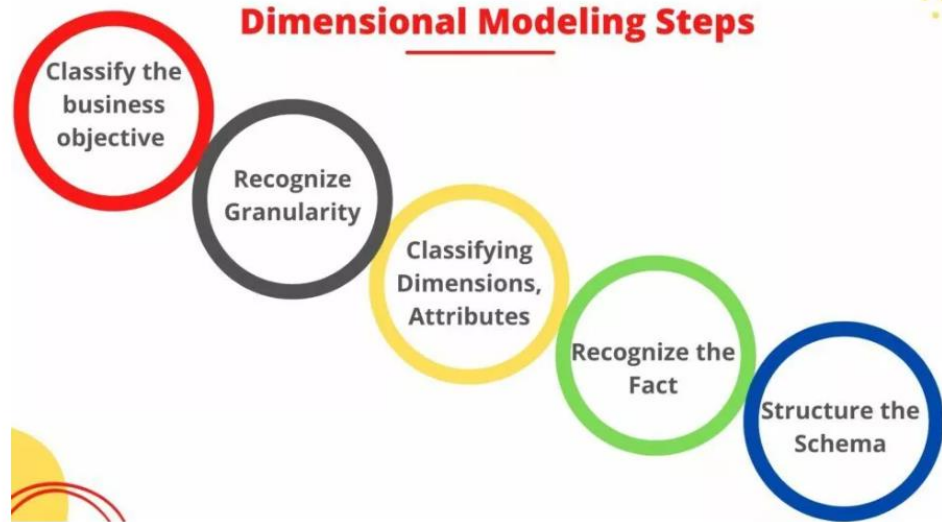
Summary

Introduction

- Dimensional modeling is a data modeling technique used in data warehousing to organize and analyze data.
- It involves designing a database schema with dimensions and facts, which are used to create a multidimensional view of data.
- The goal of dimensional modeling is to provide a flexible and efficient way to analyze large volumes of data for decision-making purposes.

Dimensional Modeling

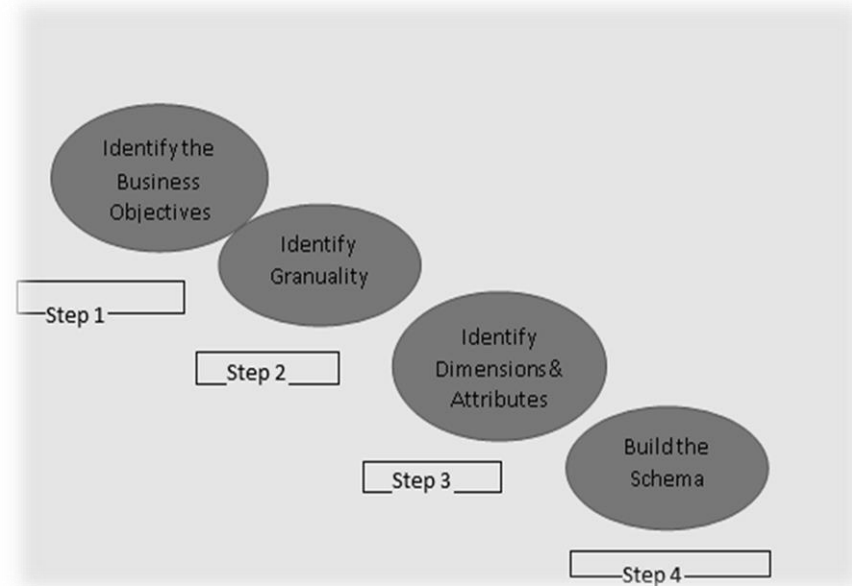
- Dimensional modeling is a data model design adopted when building a data warehouse & reduces the response time of query.



<https://www.expressanalytics.com/blog/dimensionally-speaking/>

Dimensional Modeling

- The following are the steps in Dimension modeling as shown in figure
 - Identify Business Process
 - Identify Grain (level of detail)
 - Identify dimensions and attributes
 - Build Schema



Dimensional Modeling

- **Step 1: Identify the Business Objectives**
 - Selection of the right business process to build a data warehouse and identifying the business objectives is the first step in dimension modeling. This is very important step otherwise this can lead to repeated process and software defects.
- **Step 2: Identifying Granularity**
 - The grain literally means each minute detail of the business problem. This is decomposing of the large and complex problem into the lowest level information.

Dimensional Modeling

- **Step 3: Identifying Dimensions and attributes**
 - The dimensions of the data warehouse can be understood by the entities of the database. like, items, products, date, stocks, time etc.
- **Step 4: Build the Schema the dimensions**
 - The database structure or arrangement of columns in a database table, decides the schema.
 - There are various popular schemas like, star, snowflake, fact constellation schemas - summarizing, from the selection of business process to identifying each and every finest level of detail of the business transactions.

Dimensional Modeling

- **Strengths of Dimensional Modeling**

- It provides the simplicity of architecture or schema to understand and handle various stakeholders from warehouse designers to business clients.
- It reduces the number of relationships between different data elements.
- It promotes data quality by enforcing foreign key constraints as a form of referential integrity check on a data warehouse.
- The aggregate functions used in the schemas optimize the query performance posted by the customers.

Identifying Facts & Dimensions

- **Facts and Fact table**

- A fact is an event. It is a measure which represents business items or transactions of items having association and context data.
- The Fact table contains the description of all the primary keys of all the tables used in the business processes which acts as a foreign key in the fact table.
- It also has an aggregate function to compute the business process on some entity. It is a numeric attribute of a fact, representing the performance or behavior of the business relative to the dimensions.

Identifying Facts & Dimensions

- **Dimensions and Dimension table**
 - It is a collection of data which describe one business dimension. Dimensions decide the contextual background for the facts, and they are the framework over which OLAP is performed.
 - Dimension tables establish the context of the facts. The table stores fields that describe the facts. The data in the table are in de normalized form. So, it contains large number of columns as compared to fact table.

Identifying Facts & Dimensions

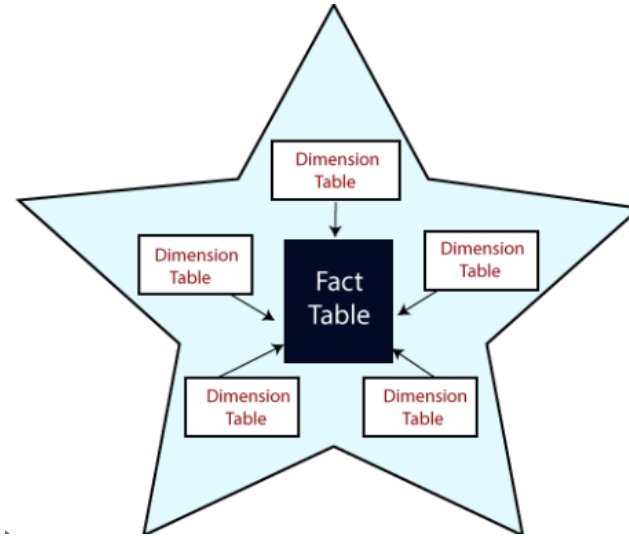
- **Student registration**
 - Fact Table (student_id, course_id, program_id, date_of_registration, fee_id)
 - Measure: Sum (Fee_amount))
 - Dimension Tables (Student_details, Course_details Program_details, Fee_details, Date)

Star Schema

- There are two basic popular models which are used for dimensional modeling:

Star Model

Snowflake Model

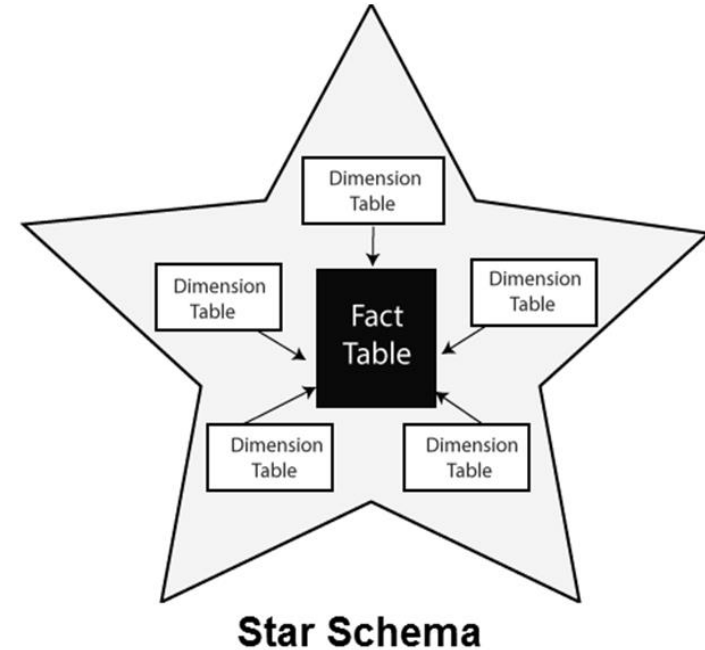


<https://www.javatpoint.com/data-warehouse-what-is-star-schema>

Star Schema

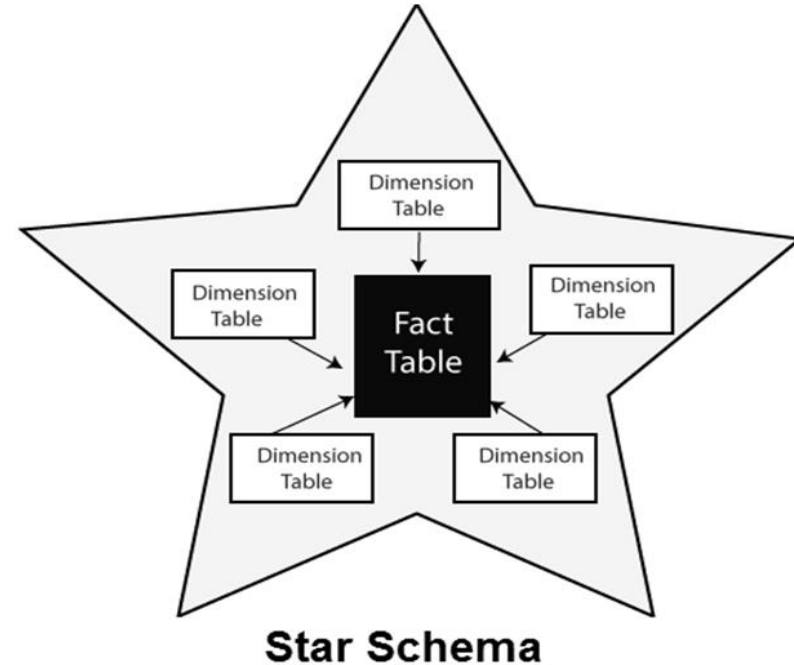
- **Star Model**

- It represents the multidimensional model. In this model the data is organized into facts and dimensions. The star model is the underlying structure for a dimensional model. It has one broad central table and a set of smaller tables arranged in a star design.



Star Schema

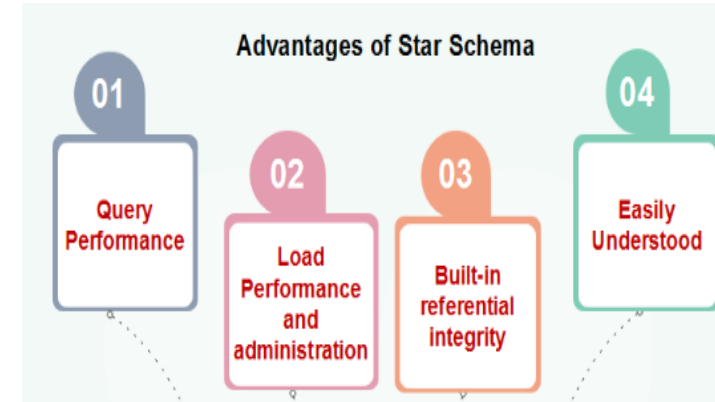
- **Features of Star Schema**
 - The data is in denormalized database.
 - It provides quick query response
 - Star schema is flexible can be changed or added easily.
 - It reduces the complexity of metadata for developers and end users.



Advantages & Disadvantages of Star Schema

- **Advantages of Star Schema**

- Query performance
- Load performance and administration
- Built-in referential integrity
- Easily understood



<https://www.javatpoint.com/data-warehouse-what-is-star-schema>

Advantages & Disadvantages of Star Schema

Disadvantages of Star Schema

- Decreased data integrity
- Less capable of handling, diverse and complex queries
- Non Many-to-Many Relationship

Snowflake Schema

- Snowflake schema is the extension of star schema which adds more dimensions to give more meaning to the logical view of the database.
- These additional tables are more normalized than star schema.

Snowflake Schema

- **Snowflake Model**

- The snowflake model is the conclusion of decomposing one or more of the dimensions.
- Snowflake Schema in data warehouse is a logical arrangement of tables in a multidimensional database such that the ER diagram resembles a snowflake shape.
- A Snowflake Schema is an extension of a Star Schema, and it adds additional dimensions. The dimension tables are normalized which splits data into additional tables.

Snowflake Schema

- **Features of snowflake schema**
 - Following are the important features of snowflake schema:
 - It has normalized tables.
 - Occupy less disk space.
 - It requires more lookup time as many tables are interconnected & extending dimensions.

Advantages of Snowflake Schema

- A Snowflake schema occupies a much smaller amount of disk space compared to the Star schema.
- Snowflake schema of small protection from various Data integrity issues.
- Most people tend to prefer the Snowflake schema because of how safe it is.
- Data is easy to maintain and more structured.

Disadvantages of Snowflake Schema

- **Complex Data Schemas:**
 - Snowflake schemas create many levels of complexity while normalizing the attributes of a star schema. This complexity results in more complicated source query joins. In offering a more efficient way to store data, snowflake can result in performance declines while browsing these complex joins.
- **Slower at processing cube data:**
 - In a snowflake schema, the complex joins result in slower cube data processing. The star schema is generally better for cube data processing.

Disadvantages of Snowflake Schema

- **Lower data integrity levels:**
 - Snowflake schemas offer greater normalization and fewer risks of data corruption after performing UPDATE and INSERT commands, they do not provide the level of transnational assurance that comes with a traditional, highly-normalized database structure.

Star Schema Vs Snowflake Schema

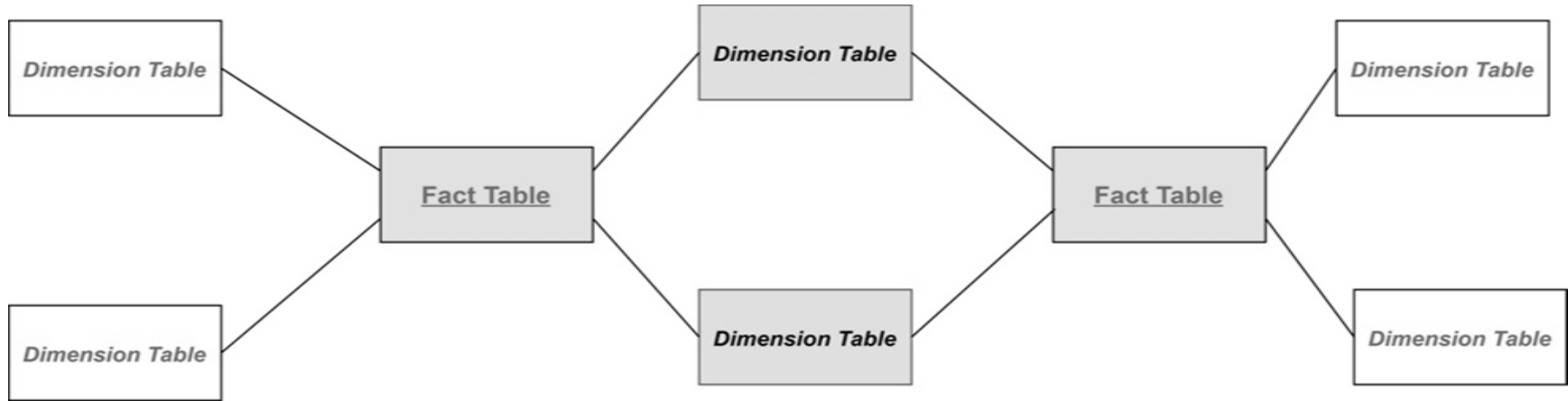
Features	Star Schema	Snowflake Schema
Normalized Dimension Tables	The dimension tables in star schema are not normalized so they may contain redundancies	This schema has normalized dimension tables
Queries	The execution of queries is relatively faster as there are less joins needed in forming a query.	The execution of snowflake schema complex queries is slower than star schema as many joins and foreign key relations are needed to form a query. Thus performance is affected.
Performance	Star schema model has faster execution and response time	It has slow performance as compared to star schema

Star Schema Vs Snowflake Schema

Storage Space	This type of schema requires more storage space as compared to snowflake due to unnormalised tables.	Snowflake schema tables are easy to maintain and save storage space due to normalized tables.
Usage	Star schema is preferred when the dimension tables have lesser rows	If the dimension table contains large number of rows, snowflake schema is preferred
Type of DW	This schema is suitable for 1:1 or 1: many relationships such as data marts.	It is used for complex relationships such as many: many in enterprise Data warehouses.
Dimension Tables	Star schema has a single table for each dimension	Snowflake schema may have more than one dimension table for each dimension.

Fact Constellation Schema

- **Fact constellation** is like the galaxy of universe containing several stars. It is a collection of fact schemas having one or more-dimension tables. This is mainly used in designing complex database systems.

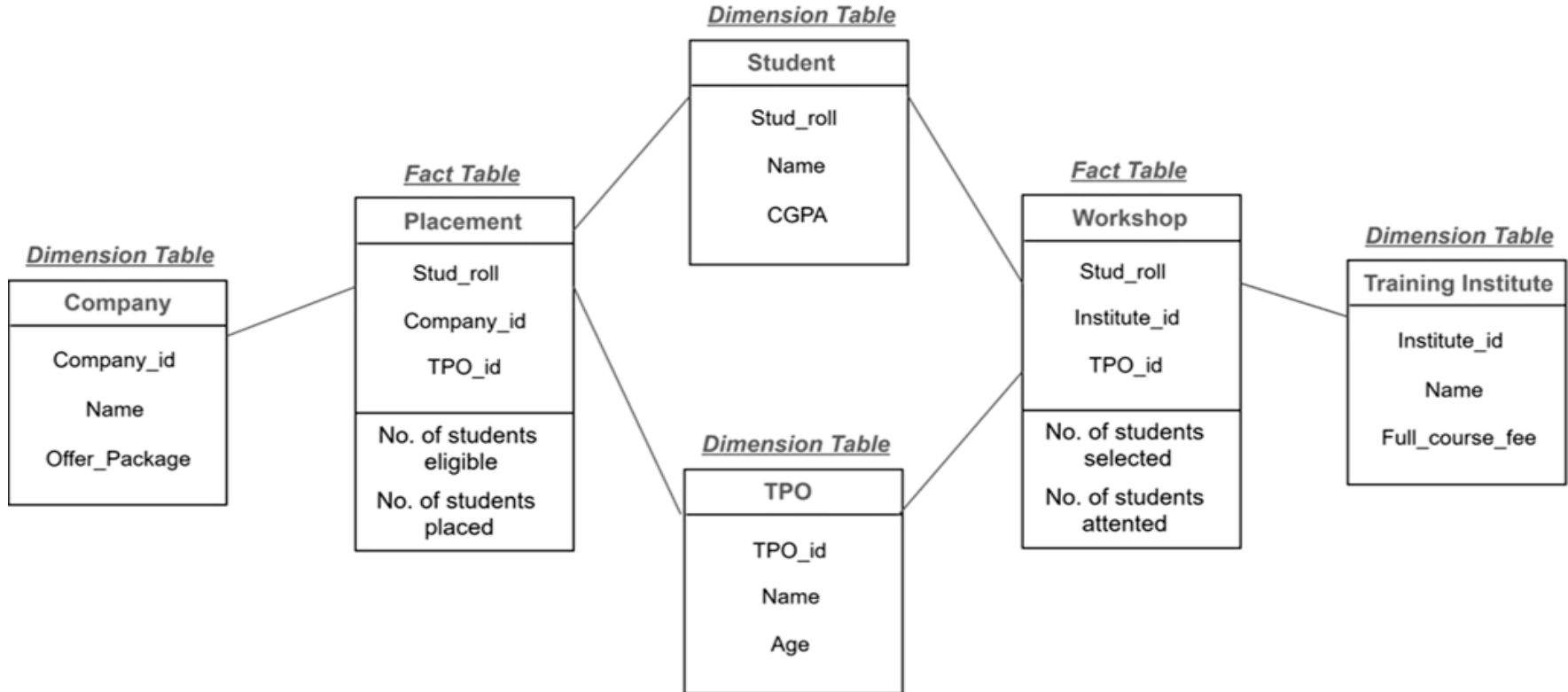


Fact Constellation Schema

- **Fact tables**

- Placement (Stud_roll, Company_id, TPO_id) , need to calculate the number of students eligible and number of students placed.
- Workshop (Stud_roll, Institute_id, TPO_id) need to find out the facts about number of students selected, number of students attended the workshop).
- So, there are two fact tables namely, Placement and Workshop which are part of two different star schemas having:
- Dimension tables – Company, Student and TPO in Star schema with fact table Placement and
- dimension tables Training Institute, Student and TPO in Star schema with fact table Workshop.

Fact Constellation Schema



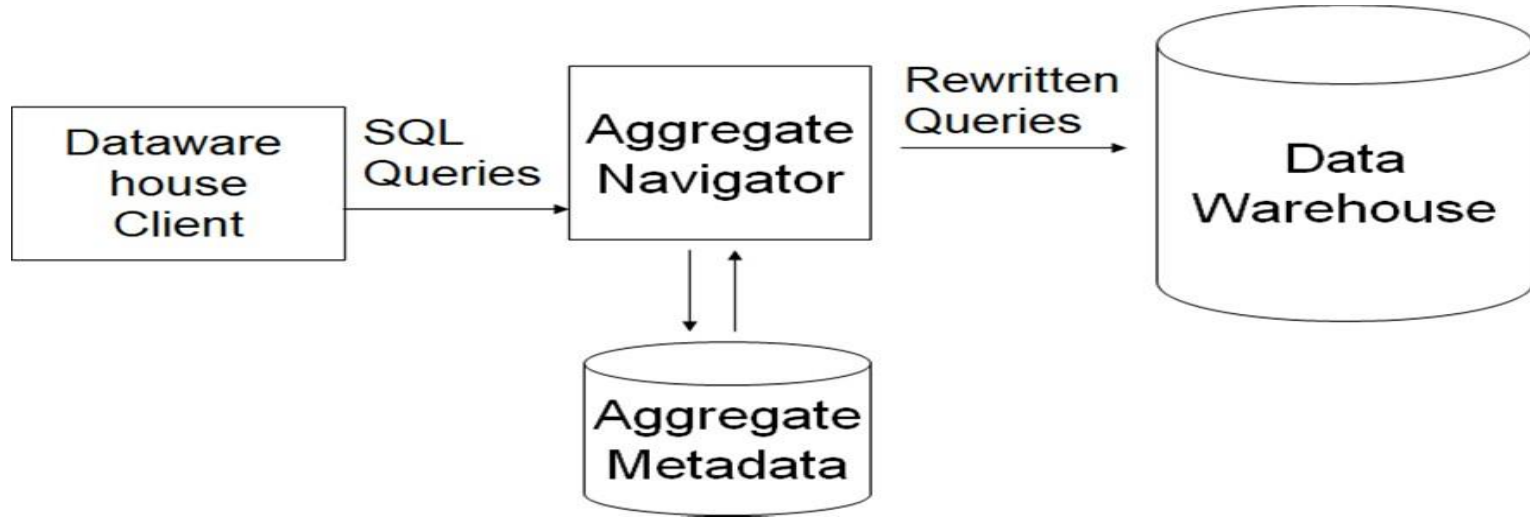
Advantages & Disadvantages of Fact Constellation Schema

- **Advantage**
 - This schema is more flexible and gives wider perspective about the data warehouse system.
- **Disadvantage**
 - Fact Constellation schema is connecting two or more facts to form a constellation. This kind of structure makes it complex to implement and maintain.

Aggregate Tables

- **Aggregate Tables**
 - Aggregate fact tables roll up the basic fact tables of the schema to improve the query processing.
 - The business tools smoothly select the level of aggregation to improve the query performance.
 - Aggregate fact tables contain foreign keys referring to dimension tables.

Aggregate Tables



Need for Building Aggregate Fact Tables

- It occupies less space than atomic fact tables. It nearly takes the half time of a general query processing.
- One of the more popular uses of aggregates is to adjust the granularity of a dimension.
- The Roll-up OLAP operation of the base fact tables generates aggregate tables.

Aggregate Fact Table and Derived Dimension Tables

- Aggregate facts are produced by calculating measures from more atomic fact tables.
- These tables contain computational SQL aggregate functions like AVERAGE, MIN, MAX, COUNT etc.
- It also contains function that helps to find output using group by.
- The aggregate fact tables produce summary statistics.
- Whenever, the speedy query handling is required the aggregate fact tables is the best option.

Aggregate Fact Table and Derived Dimension Tables

- **Fact Bill Month Receipt:** It contains aggregated receipts per month, per supplier. But the problem is it has additional foreign keys like supplier status for the month. To solve this, we have the concept of derived tables which contains additional measures and foreign keys that are not present in the base fact table.
- **Conformed Dimension** A conformed dimension is the dimension that is shared across multiple data mart or subject area. An organization may use the same dimension table across different projects without making any changes to the dimension tables.

Aggregate Fact Table and Derived Dimension Tables

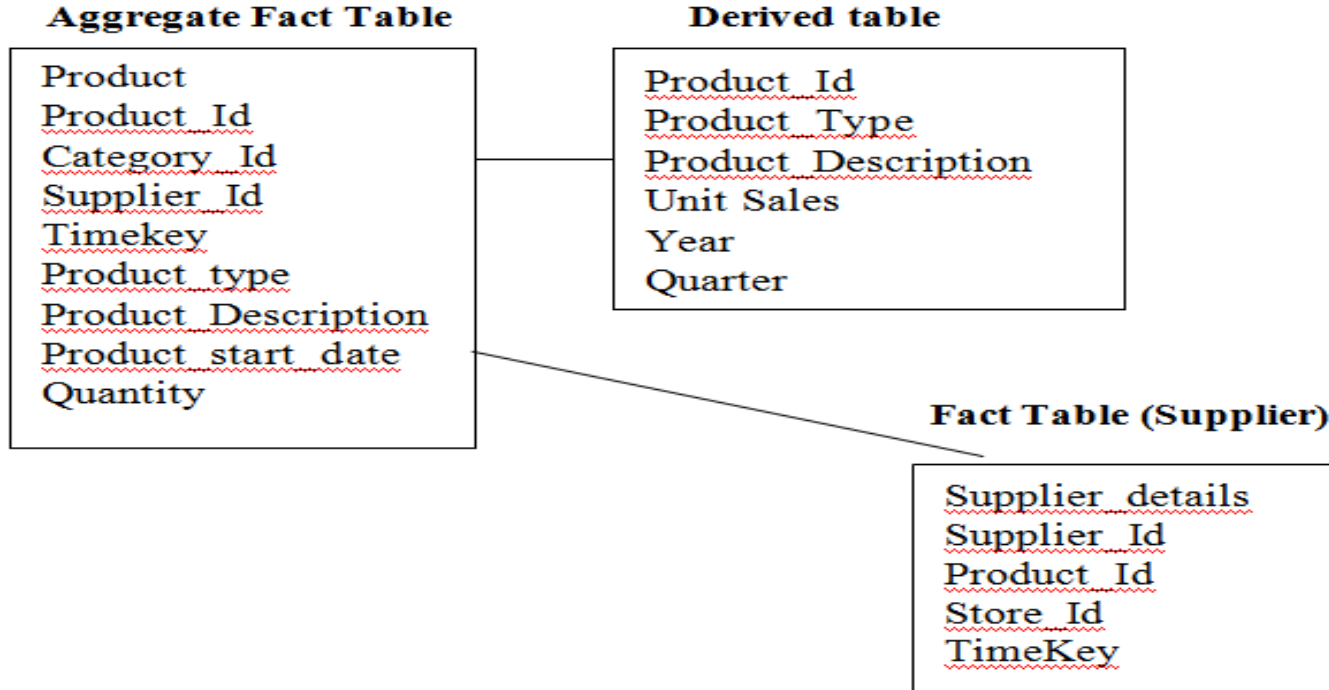
- **Derived Tables**

- It Is The Significant Addition To The Data Warehouse. Derived Tables Are Used To Create A Second-level Data Marts For Cross Functional Analysis.

- **Consolidated Fact Tables:**

- It Is The Fact Table Which Has Data From Different Fact Tables Used To Form A Schema With A Common Grain.

Aggregate Fact Table and Derived Dimension Tables



Important Topics

Dimensional modeling

Identifying facts and dimensional

Star schema and its features.

Snowflake schema and its features.

Aggregate tables.



Summary

- **In this session we have studied about :-**
- Dimensional modeling is a data model design adopted when building a data warehouse.
- Identifying facts and dimensions.
- Star model represents the multidimensional model. In this model the data is organized into facts and dimensions. The star model is the underlying structure for a dimensional model.
- Snowflake schema is the extension of star schema which adds more dimensions to give more meaning to the logical view of the database. These additional tables are more normalized than star schema.

*Thank
You*



Data Warehousing and Data Mining

Block-2 Unit-4

Extract, Transform & Loading

Topics to be Covered

Introduction

ETL and its
Need

ETL Process

Working of
ETL

ETL and
OLAP Data
Warehouses

Topics to be Covered

ETL Tools and their Benefits

Improving the Performance of ETL

ELT and its Need

Important Topics

Summary

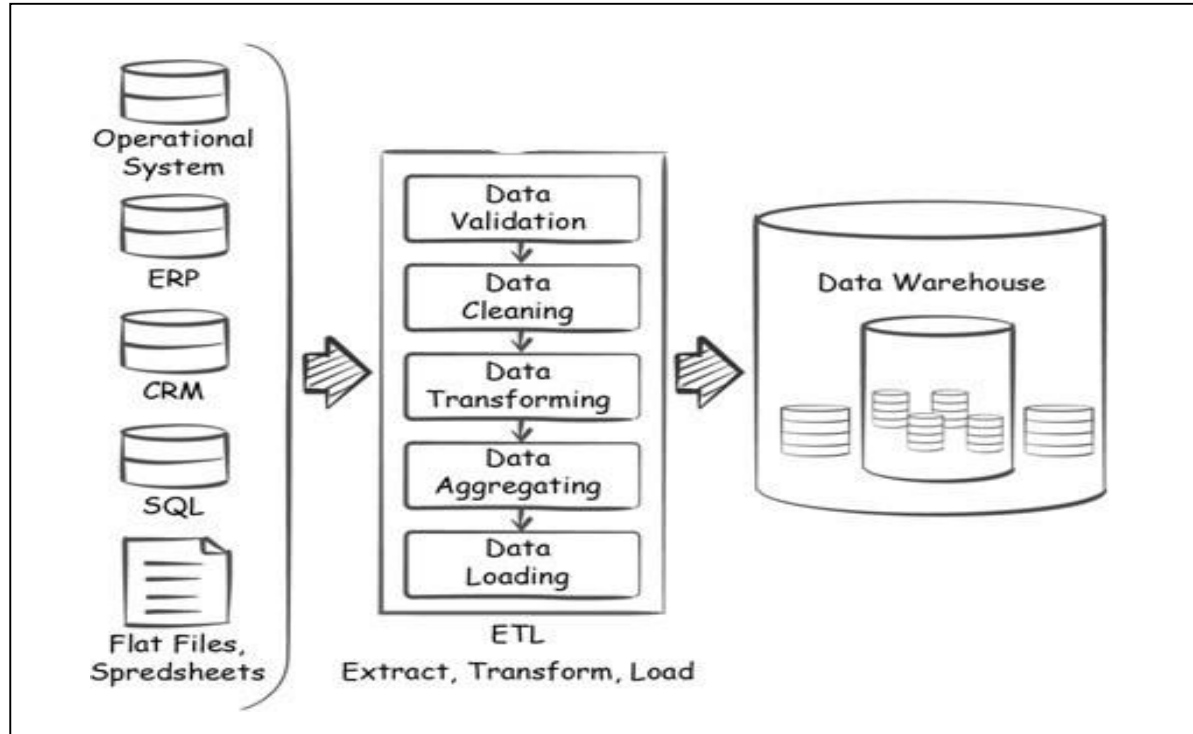
Introduction

- Extract, Transform, and Load (ETL) is a process used in data integration to collect data from various sources, transform it to fit business needs, and load it into a target database or data warehouse.
- The ETL process typically involves three phases: extraction, where data is gathered from source systems; transformation, where data is manipulated and cleaned; and loading, where data is inserted into the target system.
- ETL is a critical component in the data management and business intelligence workflows.

ETL & its Need

- Extract, Transform, Load (ETL) is a process of data integration that encompasses three steps - extraction, transformation, and loading. In a nutshell
- ETL systems take large volumes of raw data from multiple sources, convert it for analysis, and load that data into your warehouse.

ETL & its Need



ETL & its Need

- **Why Do You Need ETL?**
- Practicing ETL is also part of a healthy data management workflow, ensuring high data quality, availability, and reliability. Each of the three major components in the ETL saves time and development effort by running just once in a dedicated data flow :
 - Extract
 - Transform
 - Load

ETL Process

- ETL collects and processes data from various sources into a single data store (a data warehouse or data lake), making it much easier to analyze.
- The three steps in ETL process are mentioned below:
 - Data Extraction
 - Data Transformation
 - Data Loading

ETL Process

- **Data Extraction**

- Data extraction involves the following four steps:
 - Identify The Data To Extract
 - Estimate How Large The Data Extraction Is
 - Choose The Extraction Method
 - Assess Your SaaS Platforms

- **Data Transformation**

- Data transformation that occurs in a staging area (after extraction) is “multistage data transformation”. In ELT, data transformation that happens after loading data into the data warehouse is “in- warehouse data transformation”.

ETL Process

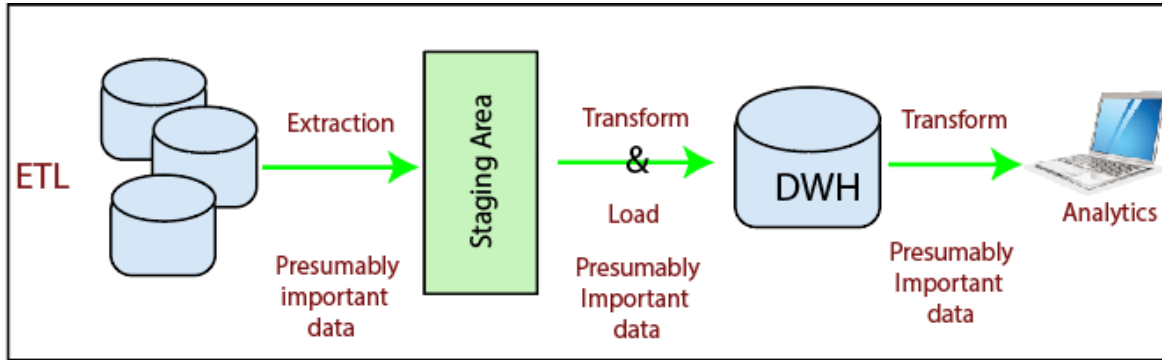
- **Data Loading**
 - Data loading is the process of loading the extracted information into your target data repository. Loading is an ongoing process that could happen through “full loading” or “incremental loading”. Types of Incremental Loads :
 - Batch incremental loads
 - Streaming incremental loads

ETL Process

- **Batch incremental loads**
 - The data warehouse ingests information in packets or batches. If it's a large batch, it's best to carry out a batch load during off-peak hours - on a daily, weekly, or monthly basis - to prevent system slowdowns .
- **Streaming incremental loads**
 - The data warehouse ingests new data as it appears in real-time. This method is particularly valuable when the end-user requires real-time updates .Further, streaming incremental loads are only possible when the updates involve a very small amount of data. In most cases, minute-by-minute batch updates offer a more robust solution than real-time streaming

ETL Process

- **Challenges in Incremental Loading**
 - Data structure changes
 - Processing data in the wrong order
 - Failure to detect problems

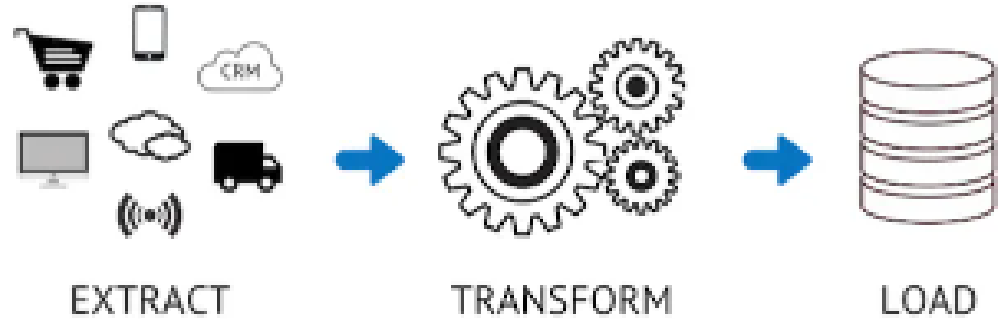


<https://www.javatpoint.com/data-warehouse-etl-vs-elt>

+91 - 9319133134 , query@cftedu.in

Working of ETL

- **ETL system performs a number of important include**
 - Parsing/Cleansing
 - Data enrichment
 - Setting velocity
 - Data validation

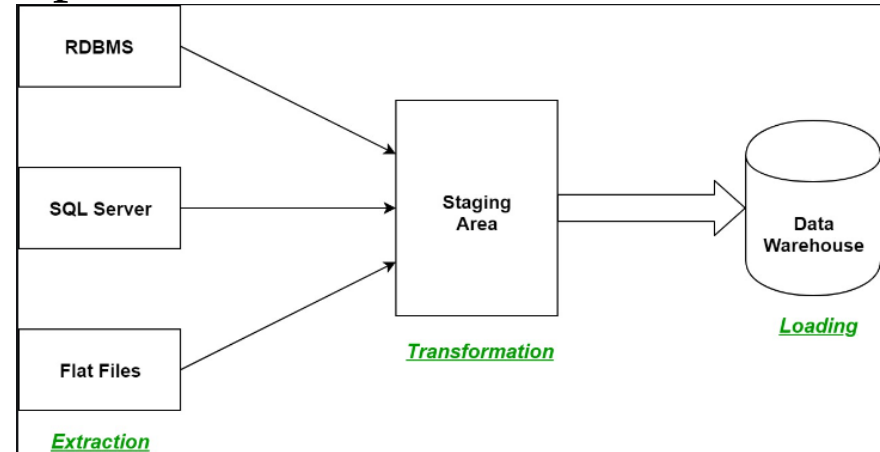


<https://www.talend.com/resources/what-is-etl/>

Working of ETL

- **Layered Implementation of ETL in a Data Warehouse**
 - ETL process is used to move data into a data warehouse, a separate layer represents each phase:

- Mirror/Raw layer
- Staging layer:
- Schema layer
- Aggregating layer



<https://www.geeksforgeeks.org/etl-process-in-data-warehouse/>

ETL & OLAP Data Warehouses

- Data engineers have been using ETL for over two decades to integrate diverse ,In business applications use online transactional processing (OLTP) database systems.
- These are optimized for writing, updating, and editing the information inside them. They are not good at reading and analysis.
- However, online analytical processing database systems are excellent at high-speed reading and analysis. That's why ETL is necessary to transform OLTP information.

ETL & OLAP Data Warehouses

- **During the ETL process, information is:**
 - Extracted from various relational database systems (OLTP or RDBMS) and other sources.
 - Transformed within a staging area, into a compatible relational format, and integrated with other data sources.
 - Loaded into the online analytical processing (OLAP) data warehouse Server.

ETL Tools & Their Benefits

ETL tools come in a wide variety both in open source or proprietary categories. There are ETL frameworks and libraries that you can use to build ETL pipelines in Python. Following are some of the benefits of the ETL tools:

- Scalability
- Simplicity
- Out-of-the-box
- Compliance
- Long-term costs

Improving the Performance of ETL

Some of the factors to be considered to improve the ETL performance:

- Tackle the Bottlenecks
- Load Data Incrementally
- Partition Large Tables
- Cut Out Extraneous Data
- Cache the Data
- Process in Parallel using Hadoop

Improving the Performance of ETL

- **Tackle the Bottlenecks:**
 - Before anything else, make sure you log metrics such as time, the number of records processed, and hardware usage.
 - Check how many resources each part of the process takes and address the heaviest one.
 - Usually, it will be the second part, building facts, and dimensions in the staging environment.

Improving the Performance of ETL

- **Load Data Incrementally**
 - Loading only the changes between the previous and the new data saves a lot of time as compared to a full load.
 - It's more difficult to implement and maintain, but difficult doesn't mean impossible, so do consider it.
 - Loading incrementally can definitely improve the ETL performance.

Improving the Performance of ETL

- **Partition Large Tables**

- In the relational databases partition large tables. That is, cut big tables down to physically smaller ones, probably by date.
- Each partition has its own indices and the indices tree is shallower thus allowing for quicker access to the data.
- It also allows switching data in and out of a table in a quick metadata operation instead of actual insertion or deletion of data records.

Improving the Performance of ETL

- **Cut Out Extraneous Data**

- It's important to collect as much data as possible, but not all of it is worthy enough to enter the data warehouse.
- To improve the ETL performance, define exactly which data should be processed and leave irrelevant rows/columns out.
- Better to start small and grow as you go as opposed to creating a giant data that takes much time to process.

Improving the Performance of ETL

- **Cache the Data**

- Caching data can greatly speed things up since memory access performs faster than do hard drives. Note that caching is limited by the maximum amount of memory your hardware supports.

- **Process in Parallel using Hadoop**

- Instead of processing serially, optimize resources by processing in parallel. Sort and aggregate functions (count, sum, etc.) block processing because they must end before the next task can begin.

ELT & Its Need

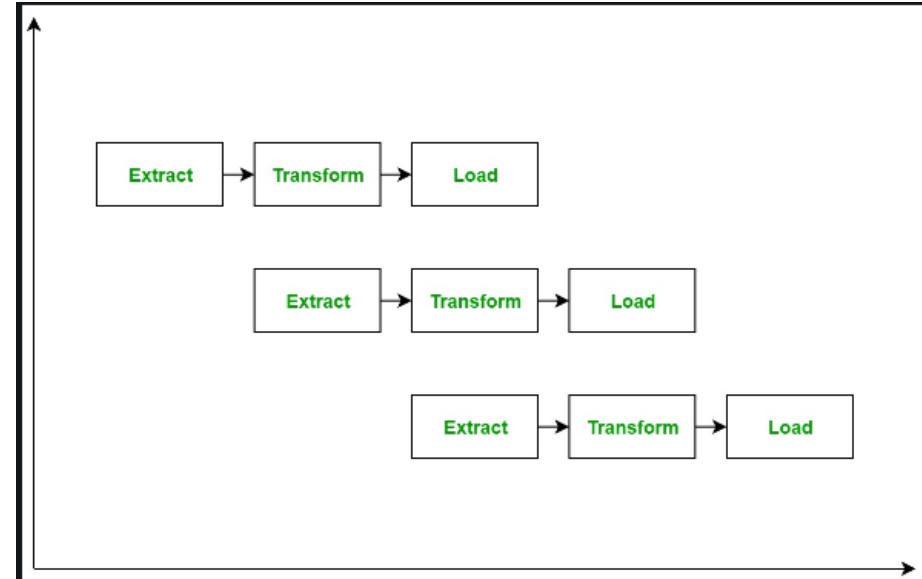
- Extract/load/transform (ELT) is the process of extracting data from one or multiple sources and loading it into a target data warehouse. ELT is an alternative to the traditional extract/transform/load (ETL) process. It pushes the transformation component of the process to the target database for better performance. This capability is very useful for processing the massive data sets needed for business intelligence (BI) and big data analytics. ELT process consists of three steps:
 - Extract
 - Load
 - Transform

ELT & Its Need

- **Extract** - This step works similarly in both ETL and ELT data management approaches. Raw streams of data from virtual infrastructure, software, and applications are ingested either in their entirety or according to predefined rules.
- **Load** – The ELT differs here with the ETL. Rather than deliver this mass of raw data and load it to an interim processing server for transformation, ELT delivers it directly to the target storage location. This shortens the cycle between extraction and delivery.

ELT & Its Need

- **Transform** - The database or data warehouse sorts and normalizes the data, keeping part or all of it on hand and accessible for customized reporting.
- The overhead for storing this much data is higher, but it offers more opportunities to mine it for relevant business intelligence in near real-time.



<https://www.geeksforgeeks.org/etl-process-in-data-warehouse/>

ELT & Its Need

ELT PROCESS



ELT & Its Need

- **Why Do You Need ELT?**
 - Large enterprises with vast data volumes.
 - Companies that require quick or frequent access to integrated data.
 - Data scientists who rely on business intelligence.
 - IT departments and data stewards interested in a low-maintenance solution.

ELT & Its Need

- **Benefits of ELT**
- The ELT process saves you steps and time. Data is first loaded into the target ecosystem, such as a data warehouse, and then transformed. Authorized users can securely access the data without returning it to source systems. No downloading is necessary for it. The ELT approach has several benefits which are listed below:
 - Get better results with more efficient effort
 - Transform your data faster
 - Combine data from different sources and formats
 - Manage data at scale
 - Save time and money

ETL Vs ELT

- The primary differences between ETL and ELT are how much data is retained in data warehouses and where data is transformed.
- With ETL, the transformation of data is done before it is loaded into a data warehouse.
- This enables analysts and business users to get the data they need faster, without building complex transformations or persistent tables in their business intelligence tools.
- Using the ELT approach, data is loaded into the warehouse as is, with no transformation before loading. This makes jobs easier to configure because it only requires an origin and a destination.

ETL Vs ELT

- The ETL and ELT approaches to data integration differ in several key ways as listed below:
 - **Load time** - It takes significantly longer to get data from source systems to the target system with ETL.
 - **Transformation time** - ELT performs data transformation on-demand, using the target system's computing power, reducing wait times for transformation.

ETL Vs ELT

- **Complexity** - ETL tools typically have an easy-to-use GUI that simplifies the process. ELT requires in-depth knowledge of BI tools, masses of raw data, and a database that can transform it effectively.
- **Data warehouse support** - ETL is a better fit for legacy on premise data warehouses and structured data. ELT is designed for the scalability of the cloud.
- **Maintenance** - ETL requires significant maintenance for updating data in the data warehouse. With ELT, data is always available in near real- time.

Important Topics

ETL Process.

Working of ETL

Difference between ETL and OLAP.

ETL tools and their Benefits.

Difference Between ELT and ETL.

Summary

- In this session we have studied and learned about :-
- The process of extracting data from source systems and bringing it into the data warehouse is commonly called ETL process, which stands for extraction, transformation, and loading.
- When an ETL process is used to move data into a data warehouse, a separate layer represents each phase.
- ETL tools come in a wide variety both in open source or proprietary categories. There are ETL frameworks and libraries that you can use to build ETL pipelines in Python.
- The primary differences between ETL and ELT are how much data is retained in data warehouses and where data is transformed.

*Thank
You*



Data Warehousing and Data Mining

Block-2 Unit-5 Part-2

Introduction to Online Analytical Processing

Topics to be Covered

Introduction

Advantages of OLAP

OLAP Architectures - MOLAP, ROLAP, HOLAP, DOLAP

Important Topics

Summary

Introduction

This session is continuation of previous session , we will see the advantages of OLAP.

OLAP is used in a variety of industries, including finance, retail, and healthcare, to perform tasks such as forecasting, trend analysis, and financial reporting.

Advantages of OLAP

Concise and Fine Data



Accessibility



Data Representation in Multi-Dimension



Business Expressions commonly used



Situational Scenarios



Easily Understood Technology

Advantages of OLAP

- **Concise and Fine Data**
 - OLAP works on the principle of combining multiple and similar records together, which are saved in multiple tables forming a schema between them as a source of connection.
 - These tables combine to form the cube to make the massive information concise and yet finely available to the user.
 - Records can be elemental right down to a single element by “drill down” and back to the cube by “drill up” operations.

Advantages of OLAP

- **Accessibility**

- The cube enables the various kinds of data like – transactional data from various resources, information about every supplier and consumer, etc. all is saved in a concise one location which is easy to operate.

- **Data Representation in Multi-Dimension**

- OLAP cube is the center of all the data. Each element of the cube contains various attributes and the number of processes performed on it.
- The cube axes are outlined by the measure and dimension of the cube which is mostly three - dimensional system.

Advantages of OLAP

- **Business Expressions commonly used**
 - The size of an OLAP cube consisting of data portrays the company's economic and financial conditions.
 - The end user does not manipulate the database files; they deal with end processes like products, salesmen, employees, customers, etc.
 - This gives a reason to even user with less to zero technical background to use LAP technology.

Advantages of OLAP

- **Situational Scenarios**

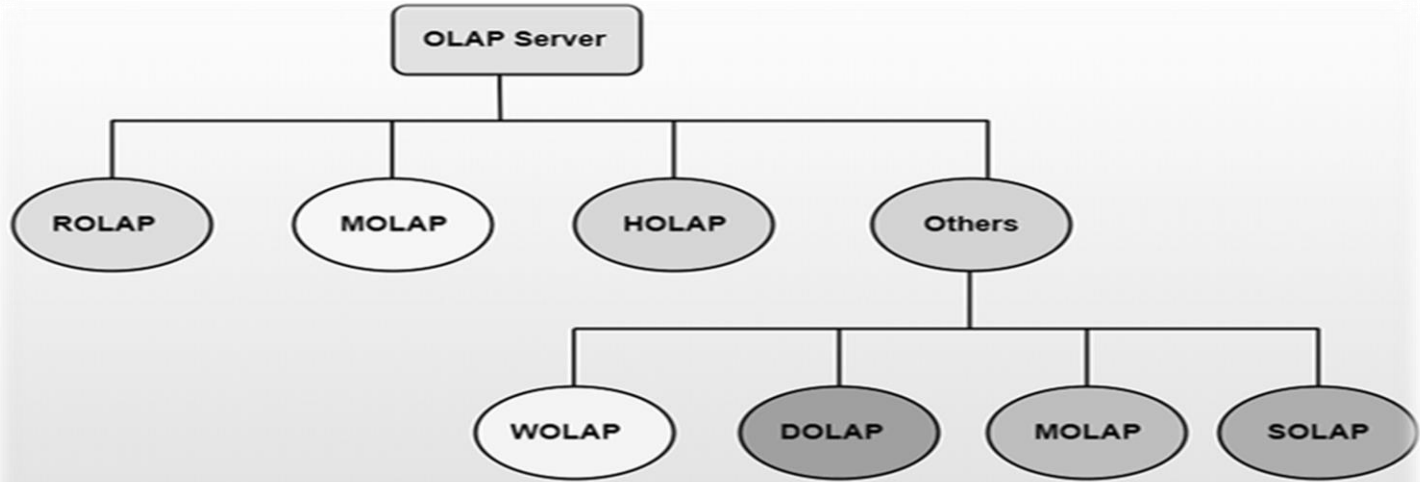
- This feature of OLAP technology is responsible for providing the customers the ability to update the values to look at the consequences brought in the cube's situation.
- Through this feature business intelligence can deeply examine the possible factors of driving a situation in a company and prevent them if necessary.

Advantages of OLAP

- **Easily Understood Technology**
 - OLAP technology providers provide their end users with enough tutorial, documents and some start off technical assistance particularly in case of web – based OLAP operations.
 - The end customers are given sessions to continuously work with a group of technical experts so that they do not have to solve all the OLAP issues by themselves.

OLAP Architecture: MOLAP, ROLAP, HOLAP & DOLAP

- There are types of OLAP Architecture:-



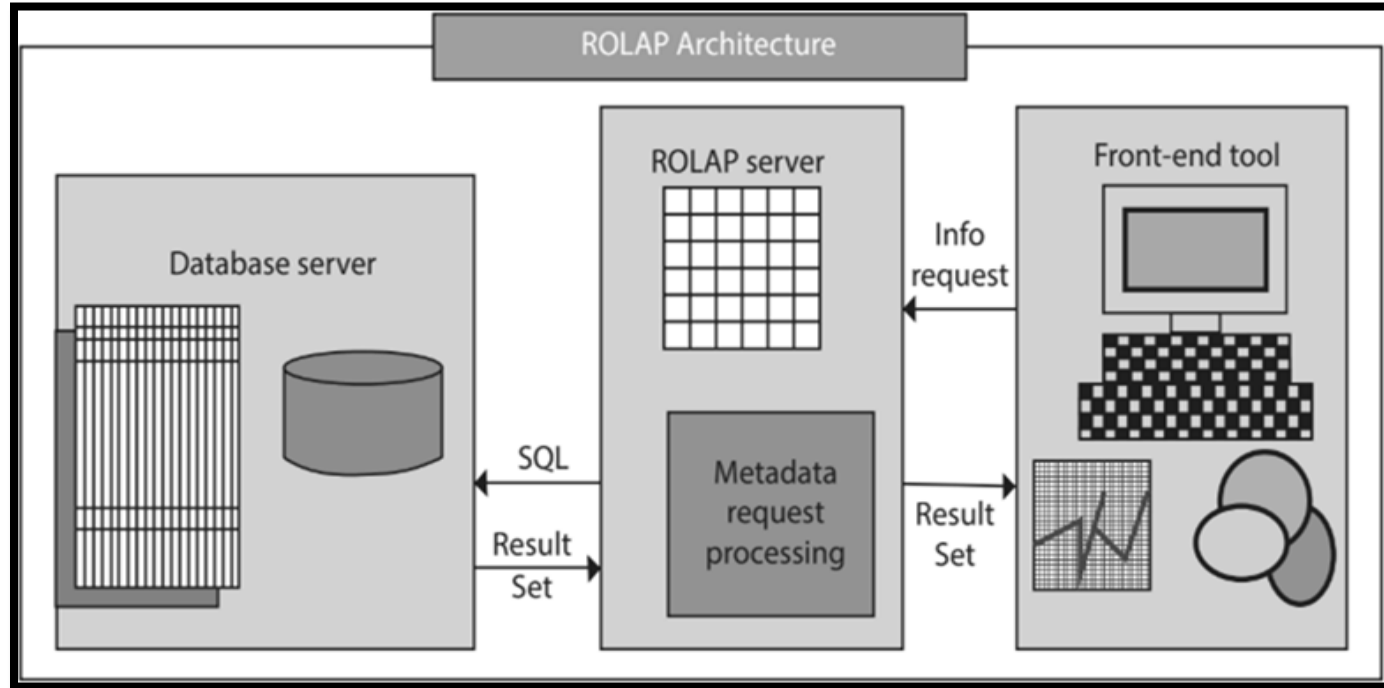
OLAP Architecture: MOLAP, ROLAP, HOLAP & DOLAP

- **ROLAP Architecture**

- ROLAP implies Relational OLAP, an application based on relational DBMS.
- It performs dynamic multidimensional analysis of data stored in a relational database. The architecture is like three-tiered. It has three components viz. front end , ROLAP server and the back end
 - Database server
 - ROLAP server
 - Front-end tool

OLAP Architecture: MOLAP, ROLAP, HOLAP & DOLAP

- **ROLAP Architecture**



OLAP Architecture: MOLAP, ROLAP, HOLAP & DOLAP

- **The characteristics of ROLAP are:**
 - ROLAP utilizes the more processing time and disk space.
 - ROLAP enables and supports larger user group in the distributed environment.
 - ROLAP processes complex queries utilizing the greater amounts of data.
 - Popular ROLAP products include Metacube by Stanford Technology Group, Red Brick Warehouse by Red Brick Systems.

OLAP Architecture: MOLAP, ROLAP, HOLAP & DOLAP

- **MOLAP Architecture**

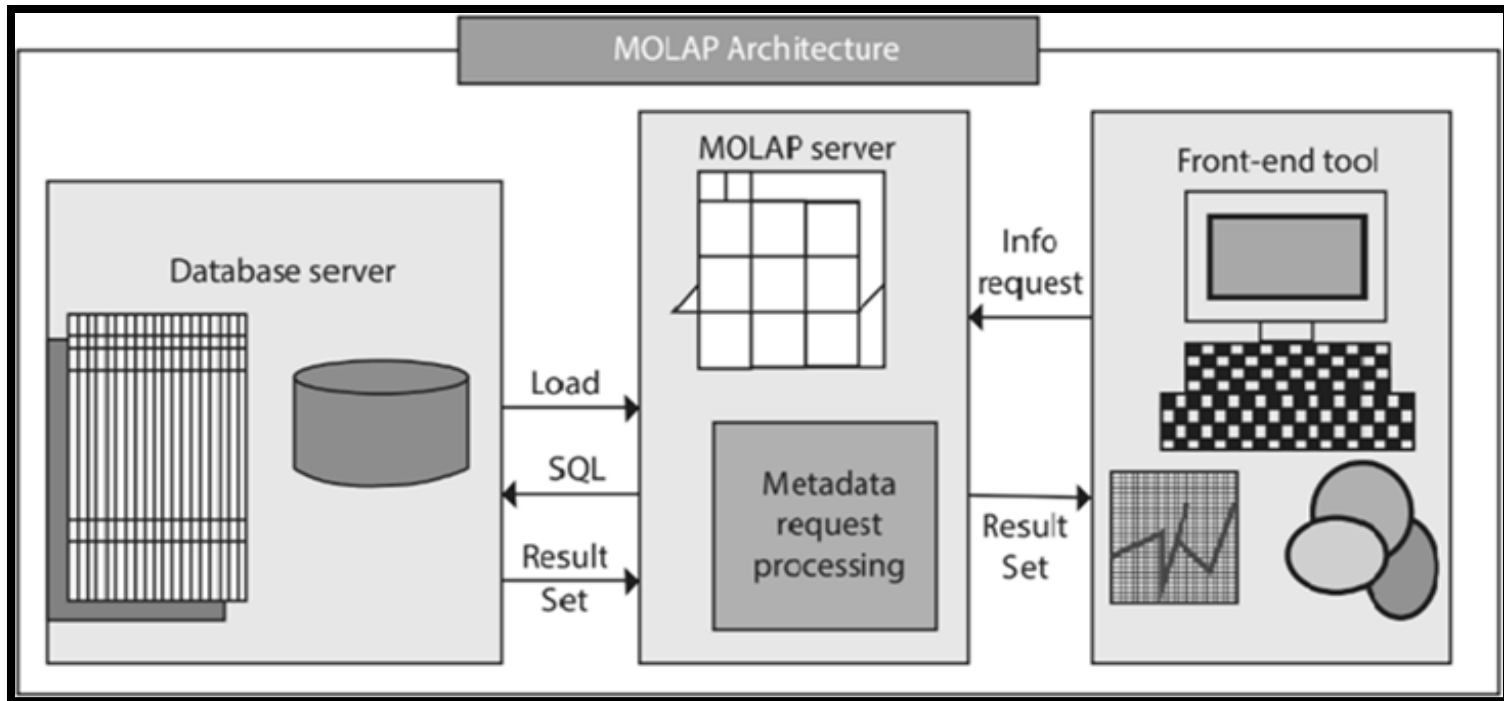
- ROLAP utilizes the more processing time and disk space.
- ROLAP enables and supports larger user group in the distributed environment.
- ROLAP processes complex queries utilizing the greater amounts of data.
- Popular ROLAP products include Metacube by Stanford Technology Group, Red Brick Warehouse by Red Brick Systems.

OLAP Architecture: MOLAP, ROLAP, HOLAP & DOLAP

- MOLAP it stands for Multidimensional Online Analytical Processing. It processes the data using the multidimensional cube using various combinations. Since, the data is stored in multidimensional structure the MOLAP engine uses the pre-computed or pre-stored and stored. The architecture has three components:
 - Database server
 - MOLAP server
 - Front-end tool

OLAP Architecture: MOLAP, ROLAP, HOLAP & DOLAP

- MOLAP Architecture**



OLAP Architecture: MOLAP, ROLAP, HOLAP & DOLAP

The Characteristics of MOLAP

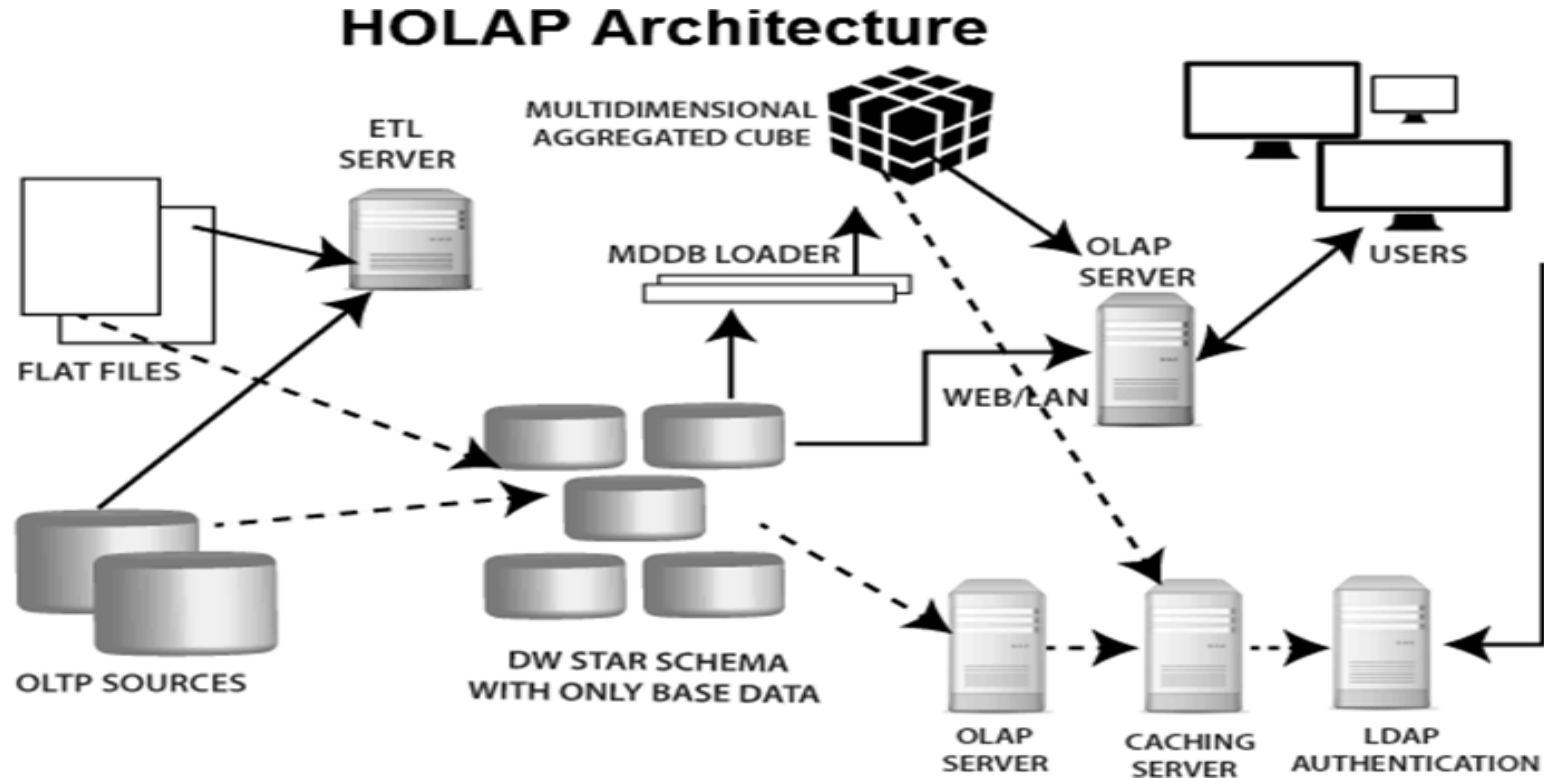
- 1.It is a user-friendly architecture, easy to use.
- 2.The OLAP operations slice and dice speeds up the data retrieval.
- 3.It has small pre-computed hypercube.

OLAP Architecture: MOLAP, ROLAP, HOLAP & DOLAP

- **HOLAP Architecture**

- It defines Hybrid Online Analytical Processing. It is the hybrid of ROLAP and MOLAP technologies. It connect both the dimensions together in one architecture.
- It stores the intermediate or part of the data in ROLAP and MOLAP. Depending on the query request it accesses the databases. It has the following components:
 - Database server
 - ROLAP and MOLAP server
 - Front-end tool

OLAP Architecture: MOLAP, ROLAP, HOLAP & DOLAP



OLAP Architecture: MOLAP, ROLAP, HOLAP & DOLAP

- **The characteristics of HOLAP are:**
 - Flexible handling of data.
 - Faster aggregation of data.
 - HOLAP can drill down the hierarchy of data and can access to relational database for any relevant and stored information in it.
 - Popular HOLAP products are Microsoft SQL Server 2000 presents a hybrid OLAP server.

OLAP Architecture: MOLAP, ROLAP, HOLAP & DOLAP

- **DOLAP Architecture**

- Desktop Online Analytical Processing (DOLAP) architecture is most suitable for local multidimensional analysis.
- It is like a miniature of multidimensional database or it's like a sub cube or any business data cube. The components are:
 - Database Server
 - DOLAP server
 - Front End

OLAP Architecture: MOLAP, ROLAP, HOLAP & DOLAP

- **The characteristics of DOLAP are:**
 - The three-tiered architecture is designed for low-end, standalone user like a small shop owner in the locality.
 - The data cube is locally stored in the system so, retrieval of results is faster.
 - No load on the backend or at the server end.
 - DOLAP is relatively cheaper to deploy

Important Topics

- Advantages of OLAP
- OLAP Architectures



<https://www.geeksforgeeks.org/olap-full-form/>

Summary

- **In this session we have seen that :-**

The concept hierarchies help to organize the dimensions into logical levels

The various OLAP operations help to extract information across sub cubes.

The creation of cube and types of OLAPs helps to understand the architecture and usage of various applications of OLAP.

*Thank
You*



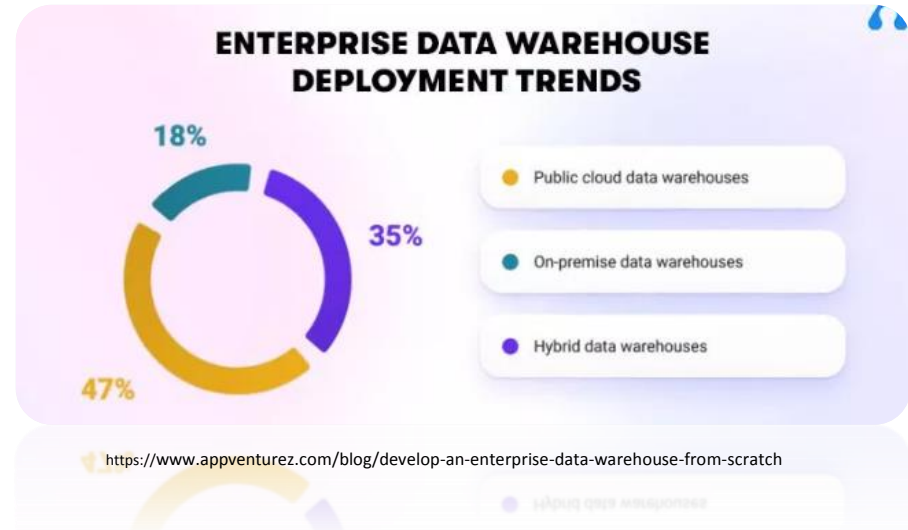
Data Warehousing and Data Mining

Block-2 Unit-6 Part -2

Trends in Data Warehouse

Topics to be Covered

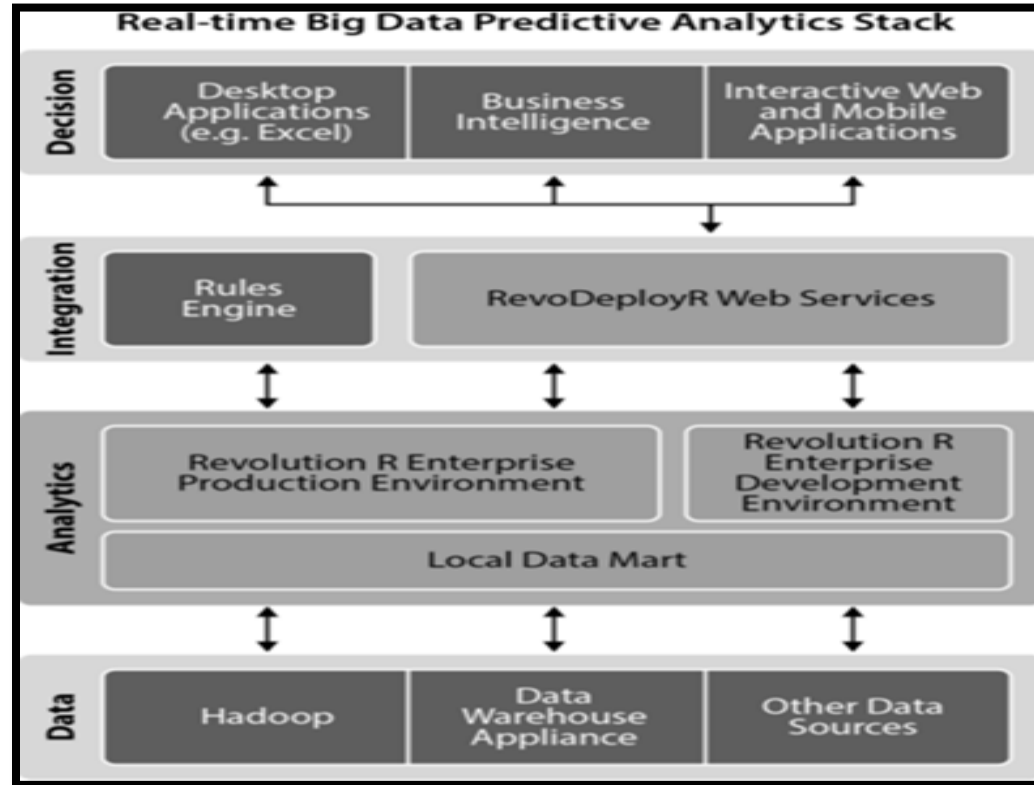
- Introduction
- Real Time Data Warehousing
- Data Warehousing and Hadoop
- Data Warehouse Automation
- Important Topics
- Summary



Real Time Data Warehousing

- The Data visualization reports need to be based on trustworthy and most current data to be of any worth to the business.
- To make this possible the underlying processes – of sourcing the data, transforming it and making it available to various functional domains across the enterprise .
- Its need to happen as optimally as possible within the shortest possible time.

Real Time Data Warehousing



Real Time Data Warehousing

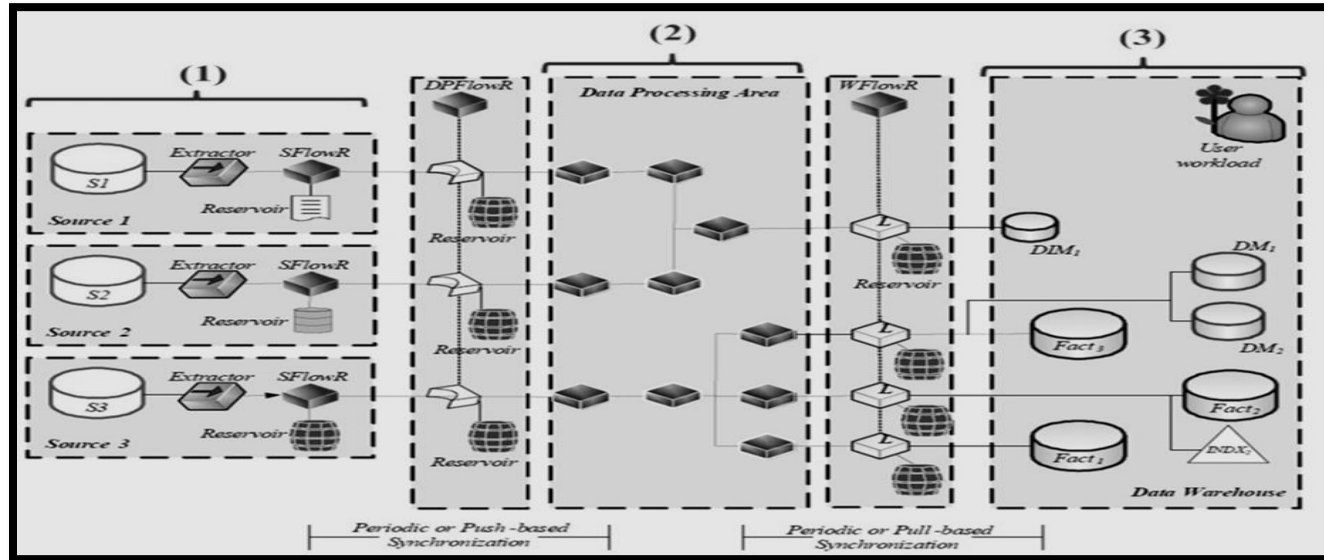
- Hadoop is a popular choice at the data-layer as it offers many tools for ingesting and transforming data in real-time. It also allows for integration of other tools like Spark for replacing the built-in MapReduce tool.
- For example, the Spark engine that powers the Shark (a RTDW) data warehouse gives a 10-100 times faster performance than MapReduce.
- The Analytics and Decision layers are bound together by the Integration Layer and this composite collection of layers is fed through various components of the data layer.

Real Time Data Warehousing

- **Real-Time Data Warehouse Architecture:-**
- The Integration Layer binds together the Analytics and Decision layers and this composite collection of layers is fed through various components of the data layer. these changes expose a typical constructs that are essential for -
 - Data Sources Are Added/Updated,
 - Data Is Propagated Across The Warehouse And
 - Data Is Loaded Into Various Repositories For Heterogeneous Data.

Real Time Data Warehousing

- Real Time Data Warehouse Architecture is shown which comprises of following component blocks:



Real Time Data Warehousing

- Data production systems hosted on data sources and responsible for populating the data warehouse.
- This block comprises of a source flow regulator (SFlowR) that periodically transmits data to the data warehouse after the data has been identified for pertinent changes.
- The data processing area (DPA) responsible for data-quality and transformation. This block contains the data flow regulator (DFlowR) used for detecting the transmission-ready source.

Real Time Data Warehousing

- **Real-Time Data Warehouse Architecture Tradeoffs :**
- It is understood from the above architectures, few changes required across the entire data warehouse ecosystem to implement a real time data warehouse. Consequently, there are a few architectural tradeoffs that users of RTDWs need to understand. Batch file replacement. The trade-offs that ensure on the RTDW journey is briefly explained as below:
 - Restrict scrutiny of data quality
 - Publish Facts with Dimensions
 - Preclude Data-Staging
 - Real-time partitions for Data Visualization
 - Real-time partition for Transactions
 - Real-time partition based on periodic snapshots

Data Warehousing & Hadoop

- **Hadoop is an Open-Source Java-Oriented framework comprising two fundamental components:**
 - Storage .
 - Processing and was developed, especially, to deal with massive volumes of data in a distributed computing ecosystem.
 - The storage component of Hadoop is called the Hadoop Distributed File System (HDFS) while the processing component is termed as MapReduce.

Data Warehousing & Hadoop

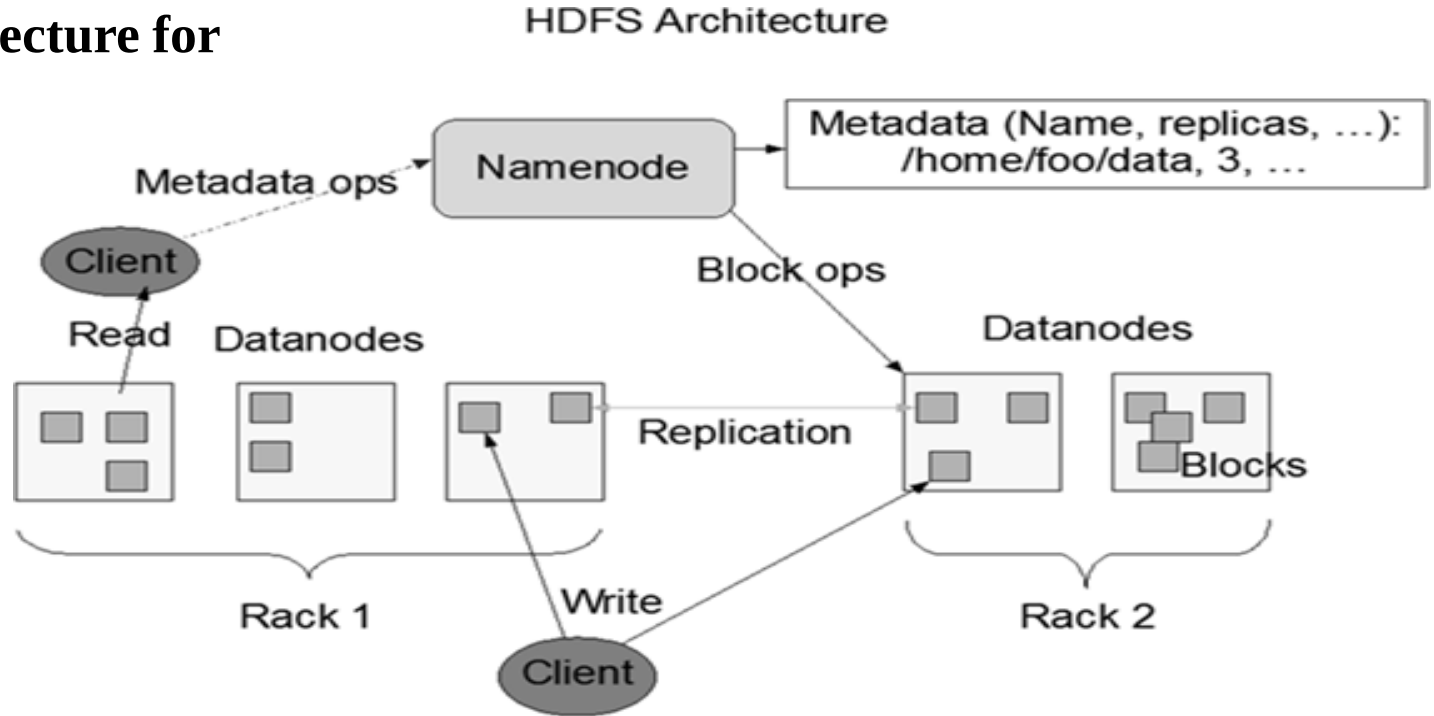
- **Essentially, Hadoop, today, comprises of the following modules:**
- **The Hadoop Distributed File System** : The HDFS allows data storage with easy accessibility. The data storage resides in an environment guaranteeing highly reliability and highly availability.
- **MapReduce** : MapReduce is principally a combination of two processes. Mapping and Reduction. Data is read from the data sources and converted to a format conducive to data analysts. This is the mapping process. The data is then subjected to mathematical operations like aggregations, grouping, encoding ranges, etc. also called as reductions.

Data Warehousing & Hadoop

- **Hadoop Common** : Hadoop common is a collection of libraries and tools required by other Hadoop modules.
- **YARN** : YARN is a platform that orchestrates and manages clustered computing resources and uses them for scheduling user-applications.
- **Hadoop Ozone** : Ozone is an object store (a data storage architecture that manages data as objects, through a combination of a globally unique identifier, metadata and the data itself).

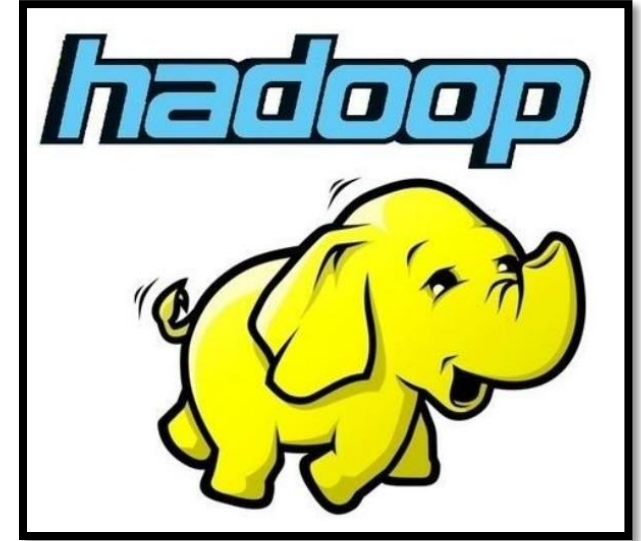
Data Warehousing & Hadoop

- **HDFS Architecture for Hadoop :**



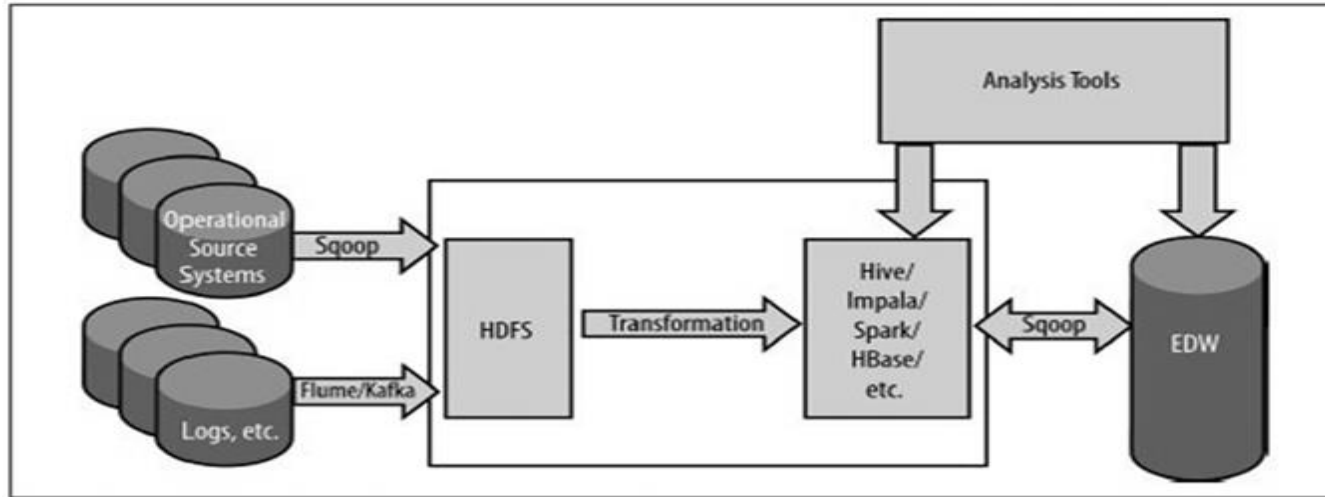
Data Warehousing & Hadoop

- The HDFS architecture as shown in Figure implements a master/slave model and has the following components:
- NameNode
- DataNodes
- The File System Namespace
- Data Replication



Data Warehousing & Hadoop

- Conceptual architecture of Hadoop Data Warehouse



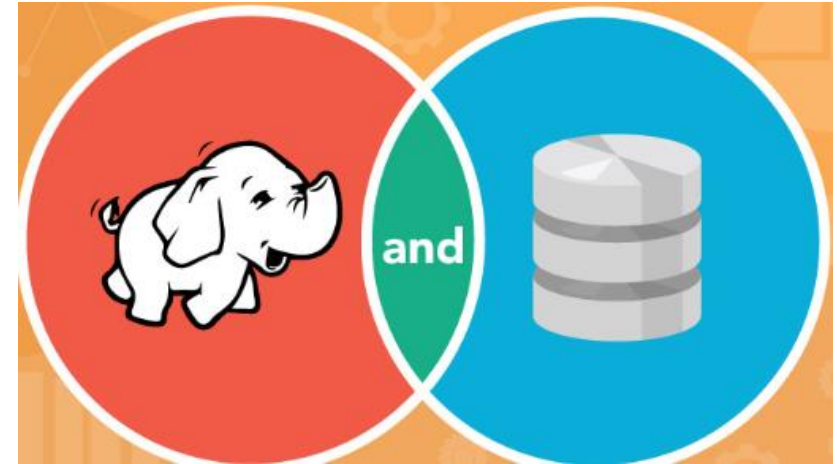
Conceptual Hadoop Data warehouse Architecture

Data Warehousing & Hadoop

- **Conceptual architecture of Hadoop Data Warehouse**
 - A Hadoop Data warehouse can be implemented using Apache Hive or Cloudera Impala. Impala offers better performance than Hive due to its massively parallel processing (MPP) capabilities.
 - The architecture in Figure10 shows a Hadoop supplemented and supported DW. In this scenario, the DW inherits benefits of the Hadoop system augmenting the ETL processes that are at the core of a DW.

Data Warehousing & Hadoop

- The components of DW in the context of Hadoop are explained as follows:
 - Data Extraction
 - Data Transformation
 - Data Loading
 - Data Analysis / Visualization



<https://www.qubole.com/blog/hadoop-and-data-warehouse>

Data Warehousing & Hadoop

- **Advantages of building a Hadoop Data Warehouse**
 - Heterogeneous data processing at scale
 - Improving ETL processing throughput
 - Good candidate for near-real-time data warehousing
 - Efficient workload management
 - Hadoop offers architectural and data level flexibility through scalable and configurable components

Data Warehousing & Hadoop

- **Challenges of building a Hadoop Data Warehouse**
 - The challenges of a Hadoop based Data warehouse are as follows:
 - Permeating the benefits of patterns in raw data to the structured components of a DW.
 - Data propagation latencies may arise in certain cases where large amounts of data are made available disproportionate to the data being consumed.
 - Use of Hadoop as an archival repository for historical data may introduce costs for import/export of data if historical data is required.

Data Warehouse Automation

- The latest trend is towards data warehouse automation (DWA) which deals with accelerating and automating the data warehouse life-cycle from conceptualization to implementation.
- Some enterprises may technically perceive it as automation beginning from analysis of source systems right up to testing and documentation processes.

Data Warehouse Automation

DWA Maturity

- DWA involves the use of sophisticated tools and architectural models to automate planning, design, integration and implementation of DWs through their life-cycle reducing repetitive and time consuming tasks like source analysis, ETL scripts deployment, etc.

Data Warehouse Automation

- **Data Warehouse Automation Tools**

- A DWA tool provides a seamless experience, free from the hassles of coding, for integration and transition of diverse data from its source to a DW and other related components. The tool automates deployment of ETL scripts, batch-processing of data and demonstrates various functions like:
 - Processes for high performance ETL and reliable ELT based integration of data
 - Modeling of data at source
 - Normalized, De-normalized and data structures with multiple dimensions
 - Seamless integration with various data sources

Data Warehouse Automation

Advantages of DWA

- End-to-end data pipeline acceleration
- Automation of data capture and streaming
- Automation of data management and integration
- Optimization of data propagation paths
- Ensure automatic data flows in their entirety
- Automatic set up of target-oriented data models
- Transformation of data lakes into DWs.

Important Topics

Data lake & the need of data lake.

Data Warehouse Vs Data Lake

Data Modeling

Cloud Data Warehousing

Real time data warehouses system

Summary

- **In this session we have seen that :-**
- Data Vault is a database modeling technique offering long-term storage of historical data arriving from multiple operational systems. The model also allows coping with issues such as auditing, data traceability, data loading performance and resilience to change.
- Data warehouse automation (DWA) which deals with accelerating and automating the data warehouse life-cycle from conceptualization to implementation.

*Thank
You*



Data Warehousing and Data Mining

Block-3 Unit-7

Data Mining – An Introduction

Topics to be Covered

- Introduction
- Data Mining and its Benefits
- Purpose Of Data Mining
- How Does Data Mining Works?
- Data Mining Techniques
- Data Mining Vs Data Warehousing
- Data Mining Tools
- Applications of Data Mining
- Issues in Data Mining
- Important Topic
- Summary

Introduction

- Data Mining is a technical process by which consistent patterns are identified, explored, sorted and organized. Data Mining is used to provide two primary advantages –
 - to give businesses the predictive power to estimate the unknown or future values
 - to provide businesses the descriptive power by finding interesting patterns in the data.
- With data mining tools, organizations of any size can extract valuable insights from their datasets, including information about consumers, costs, and future trends.

Data Mining and its Benefits

- Data mining which is also known as knowledge discovery in data (KDD) is the process of uncovering patterns and other valuable information from large data sets.
- Data mining has improved organizational decision-making through insightful data analyses. The data mining techniques that underpin these analyses can be divided into two main purposes :
 - they can either describe the target dataset or
 - they can predict outcomes through the use of machine learning algorithms.

Data Mining and its Benefits

- **The purpose of mining the data can be multi-fold which includes:**
 - Predicting various outcomes
 - Modeling target audience
 - Collecting the information about the product
- **Benefits of Data Mining**
 - It helps companies gather reliable information.
 - It's an efficient, cost-effective solution compared to other data applications.
 - It helps businesses make profitable production and operational adjustments.
 - Data mining uses both new and legacy systems.

Data Mining and its Benefits

- **Types of Data that can be Mined**

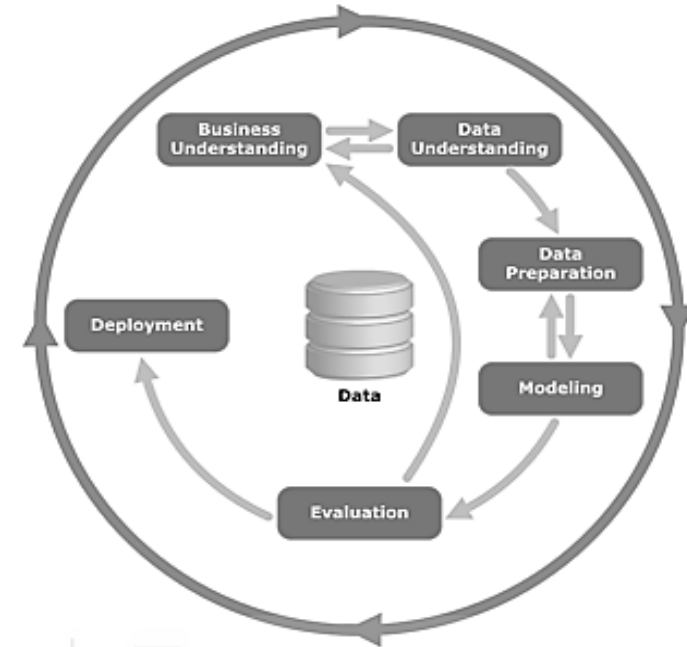
- Data mining should be applicable to any kind of information repository. However, algorithms and approaches may differ when applied to different types of data.
- Data mining is being put into use and studied for databases, including relational databases, object-relational databases and object-oriented databases, data warehouses, transactional databases, unstructured and semi-structured repositories such as the World Wide Web, social media data, advanced databases such as spatial databases, multimedia databases, time-series databases and textual databases, and even flat files

How Does Data Mining Works ?

- Cross-Industry Standard Process for Data Mining (CRISP-DM), is an industry-proven way to guide your data mining efforts. As a methodology, it includes descriptions of the typical phases of a project, the tasks involved with each phase, and an explanation of the relationships between these tasks. As a process model, it provides an overview of the data mining life cycle.
- The CRISP-DM model is flexible and can be customized easily. The life cycle model consists of six phases with arrows indicating the most important and frequent dependencies between phases.

How Does Data Mining Works ?

- Let's take a closer look at each phase of the CRISP-DM as shown in the Figure
- **Data Understanding**
 - Once the overall objective is determined, proper data needs to be collected.
 - The data must be relevant to subject matter and usually comes from a variety of sources such as sales records, customer surveys, and geo-location data.



Data Mining Life Cycle

How Does Data Mining Works ?

- **Data Preparation**

- The most time-consuming phase, the preparation phase, consists of three steps: extraction, transformation, and loading — also referred to as ETL. First, data is extracted from various sources and deposited into a staging area.
- Next, during the transformation step: the data is cleaned, null sets are populated, duplicative data is removed, errors are resolved, and all data is allocated into tables. In the final step, loading, the formatted data is loaded into the database for use.

How Does Data Mining Works ?

- **Modeling:** Data modeling addresses the relevant data set and considers the best statistical and mathematical approach to answering the objective question(s). There are a variety of modeling techniques available, such as classification, clustering, and regression analysis etc.. It's also not uncommon to use different models on the same data to address specific objectives.
- **Evaluation:** After the models are built and tested, it's time to evaluate their efficiency in answering the question identified during the business understanding phase. This is a human driven phase, as the individual running the project must determine whether the model output sufficiently meets their objectives.

How Does Data Mining Works ?

- **Deployment**

- Once the data mining model is deemed accurate and successful in answering the objective question, it's time to put it to use.
- Deployment can occur in the form of a visual presentation or a report sharing insights.
- It also can lead to action such as generating a new sales strategy or implementing risk-reduction measures.

CLASSIFICATION OF DATA MINING SYSTEMS

- **Classification according to the type of data source mined:** this classification categorizes data mining systems according to the type of data handled such as spatial data, multimedia data, time-series data, text data, World Wide Web, etc.
- **Classification according to the data model drawn on:** this classification categorizes data mining systems based on the data model involved such as relational database, object-oriented database, data warehouse, transactional, etc.

CLASSIFICATION OF DATA MINING SYSTEMS

- **Classification according to the king of knowledge discovered :** this classification categorizes data mining systems based on the kind of knowledge discovered or data mining functionalities, such as characterization, discrimination, association, classification, clustering, etc.
- **Classification according to mining techniques used :** Data mining systems employ and provide different techniques.

Data Mining Techniques

- Data mining is most useful in identifying data patterns and deriving useful business insights from those patterns. To accomplish these tasks, data miners use a variety of techniques to generate different results. Here are five common data mining techniques:
 - **Classification Analysis** - With this technique, data points are assigned to groups, or classes, based on a specific question or problem to address. For instance, if a consumer packaged goods company wants to optimize its coupon discount strategy for a specific product, it might review inventory levels, sales data, coupon redemption rates, and consumer behavioral data in order to make the best decision possible.

Data Mining Techniques

- **Association Rule Learning –**

- This function seeks to uncover the relationships between data points; it is used to determine whether a specific action or variable has any traits that can be linked to other actions (e.g., business travelers' room choices and dining habits). A hotelier might use association rule insights to offer room upgrades or food and beverage promotions to attract additional business travelers.

- **Anomaly or Outlier –**

- Detection In addition to searching for patterns, data mining seeks to uncover unusual data within a set. Anomaly detection is the process of finding data that doesn't conform to the pattern. This process can help find instances of fraud and help retailers learn more about spikes, or declines, in the sales of certain products.

Data Mining Techniques

- **Clustering Analysis -**

- Clustering looks for similarities within a data set, separating data points that share common traits into subsets. This is similar to the classification type of analysis in that it groups data points, but, in clustering analysis, the data is not assigned to previously defined groups. Clustering is useful for defining traits within a data set, such as the segmentation of customers based on purchase behavior, need state, life stage, or likely preferences in marketing.

- **Regression Analysis –**

- Regression analysis is about understanding which factors within a data set are most important, which can be ignored, and how these factors interact.

Data Mining Vs Data Warehousing

Data Mining	Data Warehouse
Data mining is the process of analyzing unknown patterns of data.	A data warehouse is database system which is designed for analytical instead of transactional work.
Data mining is a method of comparing large amounts of data to finding right patterns.	Data warehousing is a method of centralizing data from different sources into one common repository.
Data mining is usually done by business users with the assistance of engineers.	Data warehousing is a process which needs to occur before any data mining can take place.
Data mining is the considered as a process of extracting data from large data sets.	On the other hand, Data warehousing is the process of pooling all relevant data together.
One of the most important benefits of data mining techniques is the detection and identification of errors in the system.	One of the pros of Data Warehouse is its ability to update consistently. That's why it is ideal for the business owner who wants the best and latest features.

Data Mining Vs Data Warehousing

Data Mining	Data Warehouse
Data mining helps to create suggestive patterns of important factors. Like the buying habits of customers, products, sales. So that, companies can make the necessary adjustments in operation and production.	Data Warehouse adds an extra value to operational business systems like CRM systems when the warehouse is integrated.
The Data mining techniques are never 100% accurate and may cause serious consequences in certain conditions.	In the data warehouse, there is great chance that the data which was required for analysis by the organization may not be integrated into the warehouse. It can easily lead to loss of information.
The information gathered based on Data Mining by organizations can be misused	Data warehouses are created for a huge IT project. Therefore, it involves high

Data Mining Vs Data Warehousing

Data Mining	Data Warehouse
against a group of people.	maintenance system which can impact the revenue of medium to small-scale organizations.
After successful initial queries, users may ask more complicated queries which would increase the workload.	Data Warehouse is complicated to implement and maintain.
Organizations can benefit from this analytical tool by equipping pertinent and usable knowledge-based information.	Data warehouse stores a large amount of historical data which helps users to analyze different time periods and trends for making future predictions.
Organizations need to spend lots of their resources for training and Implementation purpose. Moreover, data mining tools work in different manners due to different algorithms employed in their design.	In Data warehouse, data is pooled from multiple sources. The data needs to be cleaned and transformed. This could be a challenge.

Data Mining Vs Data Warehousing

Data Mining	Data Warehouse
The data mining methods are cost-effective and efficient compares to other statistical data applications.	Data warehouse's responsibility is to simplify every type of business data. Most of the work that will be done on user's part is inputting the raw data.
Another critical benefit of data mining techniques is the identification of errors which can lead to losses. Generated data could be used to detect a drop-in sale.	Data warehouse allows users to access critical data from the number of sources in a single place. Therefore, it saves user's time of retrieving data from multiple sources.
Data mining helps to generate actionable strategies built on data insights.	Once you input any information into Data warehouse system, you will unlikely to lose track of this data again. You need to conduct a quick search, helps you to find the right statistic information.

Data Mining Tools

- Data mining techniques are derived that utilize the domain knowledge from statistical analysis, artificial intelligence, and database systems in order to analyze data in a proper manner in view of different dimensions and perspectives.
- These are the most popular data mining tools are Orange, SaS, Rattle Data Mining, Rapidminer, DataMelt Data Mining, Oracle BI etc.

Data Mining Tools

- **Orange Data Mining**
 - Displaying data table and allowing to select features
 - Reading of data.
 - Training predictors and comparison of learning algorithms
 - Data element visualization, etc.
- Data coming to orange is formatted quickly to the desired pattern, and the widgets can be easily transferred wherever and whenever needed

Data Mining Tools

- **SAS Data Mining**

- SAS stands for **Statistical Analysis System**. It is a product of the SAS Institute created for analytics and data management.
- SAS can mine data, change it, manage information from various sources, and analyze statistics. It offers a graphical UI for non-technical users.
- SAS data miner allows users to analyze big data and provide accurate insight for timely decision-making purposes.

Data Mining Tools

- **Rattle Data Mining**

- Rattle is a popular GUI for data mining using R.
- It presents statistical and visual summaries of data, transforms data so that it can be readily modeled, builds both unsupervised and supervised machine learning models from the data, presents the performance of models graphically, and scores new datasets for deployment into production.

- **Rapid Miner**

- Rapid Miner is one of the most popular tools for performing predictions
- It is written in JAVA programming language. It offers an integrated environment for text mining, deep learning, machine learning, and predictive analysis.

Data Mining Tools

- **Data Melt**

- DataMelt is a free to use tool for numeric computation, mathematics, data analysis, and data visualization. This program offers you the simplicity of scripting languages, like Python, Ruby, Groovy with the power of hundreds of Java packages. Features:
- DataMelt offers statistics, analysis of large data volumes, and scientific visualization.
- You can use it with different programming languages on different operating systems.
- It allows you to create high-quality vector-graphics images (EPS, SVG, PDF, etc.), which can be included in LaTeX and another text processor.

Applications of Data Mining

- Data Mining is primarily used today by companies with a strong consumer focus, in the areas like retail, financial, communication, and marketing organizations, to “drill down” into their transactional data and determine pricing, customer preferences and product positioning, impact on sales, customer satisfaction and corporate profits. Following are some of the applications of data mining:
 - **Basket Analysis**
 - In its most basic application, retailers use basket analysis to analyze what consumers buy (or put in their “baskets”). This is a form of the association technique, giving retailers insight into buying habits and allowing them to recommend other purchases.

Applications of Data Mining

- **Sales Forecasting**

- Sales forecasting is a form of predictive analysis to which businesses are devoting more of their budgets. Data mining can help businesses project sales and set targets by examining historical data such as sales records, financial indicators (e.g., consumer price index, inflation markers etc.).

- **Bioinformatics**

- Data Mining approaches seem ideally suited for Bioinformatics, since it is data-rich. Mining biological data helps to extract useful knowledge from massive datasets gathered in biology, and in other related life sciences areas such as medicine and neuroscience.

Applications of Data Mining

- **Inventory Planning**

- Data mining can provide businesses with up-to-date information regarding product inventory, delivery schedules, and production requirements. Data mining also can help remove some of the uncertainty that comes with simple supply-and-demand issues within the supply chain.

- **Customer Segmentation**

- Traditional market research may help us to segment customers but data mining goes in deep and increases market effectiveness. Data mining aids in aligning the customers into a distinct segment and can tailor the needs according to the customers.

Applications of Data Mining

- **Customer Relationship**

- Management Customer Relationship Management is all about acquiring and retaining customers, also improving customers' loyalty and implementing customer focused strategies. To maintain a proper relationship with a customer a business need to collect data and analyze the information.

- **Healthcare**

- Mining can be used to predict the volume of patients in every category. Processes are developed that make sure that the patients receive appropriate care at the right place and at the right time.

Applications of Data Mining

- **Education**

- There is a new emerging field, called Educational Data Mining (EDM), concerns with developing methods that discover knowledge from data originating from educational Environments.

- **Intrusion Detection**

- Any action that will compromise the integrity and confidentiality of a resource is an intrusion. The defensive measures to avoid an intrusion includes user authentication, avoid programming errors, and information protection.

Applications of Data Mining

- **Criminal Investigation**

- Criminology is a process that aims to identify crime characteristics. Actually crime analysis includes exploring and detecting crimes and their relationships with criminals.

- **Fraud Detection**

- While frequently occurring patterns in data can provide teams with valuable Fraud Detection While frequently occurring patterns in data can provide teams with valuable.

Applications of Data Mining

- **Operational Optimization**

- Process mining leverages data mining techniques to reduce costs across operational functions, enabling organizations to run more efficiently. This practice has helped to identify costly bottlenecks and improve decision-making among business leaders.

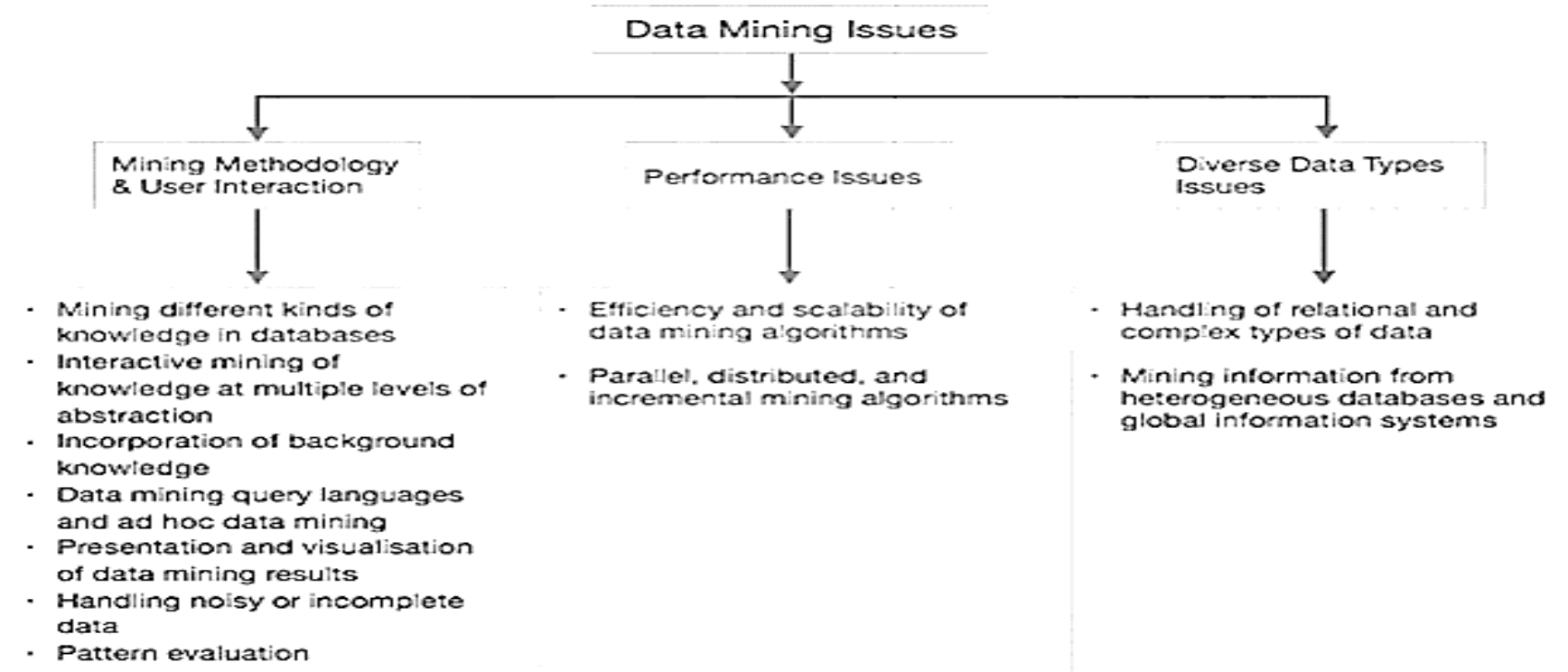


<https://www.educba.com/advantages-of-data-mining/>

Issues in Data Mining

- Data mining is not an easy task, as the algorithms used can get very complex and data is not always available at one place. It needs to be integrated from various heterogeneous data sources. These factors also create some issues. Here in this tutorial, we will discuss the major issues regarding:
 - Mining Methodology and User Interaction
 - Performance Issues
 - Diverse Data Types Issues

Issues in Data Mining



Issues in Data Mining

- **Issues in Mining Methodology and User Interaction :**
 - **Mining different kinds of knowledge in databases** – Different users may be interested in different kinds of knowledge. Therefore it is necessary for data mining to cover a broad range of knowledge discovery task.
 - **Interactive mining of knowledge at multiple levels of abstraction** – The data mining process needs to be interactive because it allows users to focus the search for patterns, providing and refining data mining requests based on the returned results.
 - **Incorporation of background knowledge** – To guide discovery process and to express the discovered patterns, the background knowledge can be used. Background knowledge may be used to express the discovered patterns not only in concise terms but at multiple levels of abstraction.

Issues in Data Mining

- **Data mining query languages and ad hoc data mining** – Data Mining Query language that allows the user to describe ad hoc mining tasks, should be integrated with a data warehouse query language and optimized for efficient and flexible data mining.
- **Presentation and visualization of data mining results** – Once the patterns are discovered it needs to be expressed in high level languages, and visual representations. These representations should be easily understandable.
- **Handling noisy or incomplete data** – The data cleaning methods are required to handle the noise and incomplete objects while mining the data regularities. If the data cleaning methods are not there then the accuracy of the discovered patterns will be poor.

Issues in Data Mining

- **Pattern evaluation** – The patterns discovered should be interesting because either they represent common knowledge or lack novelty.

- **Performance Issues**

- **Efficiency and scalability of data mining algorithms** – In order to effectively extract the information from huge amount of data in databases, data mining algorithm must be efficient and scalable.
- **Parallel, distributed, and incremental mining algorithms** – The factors such as huge size of databases, wide distribution of data, and complexity of data mining methods motivate the development of parallel and distributed data mining algorithms. These algorithms divide the data into partitions which is further processed in a parallel fashion.

Issues in Data Mining

- **Diverse Data Types Issues**

- **Handling of relational and complex types of data** – The database may contain complex data objects, multimedia data objects, spatial data, temporal data etc. It is not possible for one system to mine all these kind of data.
- **Mining information from heterogeneous databases and global information systems** – The data is available at different data sources on LAN or WAN. These data source may be structured, semi structured or unstructured. Therefore mining the knowledge from them adds challenges to data mining.

Important Topics

Data Mining and its Benefits

Purpose Of Data Mining

Working of Data Mining

Data Mining Techniques

Data Mining Vs Data Warehousing

Data Mining Tools Applications of Data Mining Issues in Data Mining

Summary

- **In this session we have studied about :-**
- Data mining is a technique that discovers previously unknown relationships in data.
- It is also known as Knowledge Discovery in Data (KDD). It is the practice of automatically searching large stores of data to discover patterns and trends that go beyond simple analysis.
- It uses sophisticated mathematical algorithms to segment the data and to predict the likelihood of future events based on past events.
- The key properties of data mining are namely - automatic discovery of patterns, prediction of likely outcomes, creation of actionable information and focus on large data sets and databases.

*Thank
You*



Data Warehousing and Data Mining

Block-3 Unit-8

Data Preprocessing

Topics to be Covered

- Introduction
- Data Preprocessing – An Overview
- Data Preprocessing Stages
- Data Cleaning
- Data Integration
- Data Transformation
- Data Reduction
- Important Topics
- Summary



Introduction

- In this unit, we will study fundamental step in the data mining, known as data preprocessing.
- Data preprocessing is the process of transforming raw data into an understandable format. It is also an important step in data mining as we cannot work with raw data.
- The quality of the data should be checked before applying machine learning or data mining algorithms. Preprocessing of data is mainly to check the data quality.

Data Preprocessing – An Overview

- Data preprocessing is the process of transforming raw data into a useful, understandable format.
- Real-world or raw data usually has inconsistent formatting, human errors, and can also be incomplete.
- Data preprocessing resolves such issues and makes datasets more complete and efficient to perform data analysis.
- Data preprocessing is transforming data into a form that computers can easily work on.
- It makes data analysis or visualization easier and increases the accuracy and speed of the machine learning algorithms that train on the data.

Data Preprocessing – An Overview

- **Purpose of Data Preprocessing**

- Important attributes cannot always be usable.
- Valid data cannot be recorded because of misunderstanding or equipment failure.
- Data that disputed other recorded data should have been discarded.
- Missing data may be inferred, especially for tuples with missing values for certain attributes.
- Data collection techniques may be unreliable.
- At the moment of data entry, human or computer errors may have existed.
- Data processing failures can also occur.

Data Preprocessing – An Overview

- **Factors Contributing to Data Quality**

- **Accuracy:** Accuracy means that the information is correct. Outdated information, typos, and redundancies can affect a dataset's accuracy.
- **Consistency:** The data should have no contradictions. Inconsistent data may give you different answers to the same question.
- **Completeness:** The dataset shouldn't have incomplete fields or lack empty fields. This characteristic allows data scientists to perform accurate analyses as they have access to a complete picture of the situation the data describes.
- **Validity:** A dataset is considered valid if the data samples appear in the correct format, are within a specified range, and are of the right type. Invalid datasets are hard to organize and analyze.

Data Preprocessing – An Overview

Timeliness: Data should be collected as soon as the event it represents occurs.

As time passes, every dataset becomes less accurate and useful as it doesn't represent the current reality.

Therefore, the topicality and relevance of data is a critical data quality characteristic.

-
- ```

graph TD
 A[Data Integration] --- B[Data Transformation]
 B --- C[Data Reduction]
 C --- D[Data Cleaning]
 D --- A
 E[Data Preprocessing]

```
- The diagram illustrates the five steps of Data Preprocessing in a circular flow. The steps are: Data Integration, Data Transformation, Data Reduction, Data Cleaning, and Data Preprocessing (centered). The steps are connected by a circular line, indicating a continuous process.

# Data Preprocessing Stages

- **Data Cleaning**

- Data cleaning or cleansing is the process of cleaning datasets by accounting for missing values, removing outliers, correcting inconsistent data points, and smoothing noisy data.
- In essence, the motive behind data cleaning is to offer complete and accurate samples for machine learning models.
- The techniques used in data cleaning are specific to the data scientist's preferences and the problem they're trying to solve.
- Missing values and noisy data are issues that are solved during data cleaning and the techniques involved.

# Data Preprocessing Stages

- **Data Integration**

- As data is collected from various sources, data integration is a crucial part of data preparation. Integration may lead to several inconsistent and redundant data points, ultimately leading to models with inferior accuracy. Following are some approaches to integrate data:
- **Data consolidation:** Data is physically brought together and stored in a single place. Having all data in one place increases efficiency and productivity. This step typically involves using data warehouse software.
- **Data virtualization:** In this approach, an interface provides a unified and real-time view of data from multiple sources. In other words, data can be viewed from a single point of view.

# Data Preprocessing Stages

- **Data propagation:** Involves copying data from one location to another with the help of specific applications. This process can be synchronous or asynchronous and is usually event-driven.
- **Data Reduction**
  - Data reduction is used to reduce the amount of data and thereby reduce the costs associated with data mining or data analysis. It offers a condensed representation of the dataset. Although this step reduces the volume, it maintains the integrity of the original data.

# Data Preprocessing Stages

- **Data Transformation**

- Mining Data Transformation is the process of converting data from one format to another. In essence, it involves methods for transforming data into appropriate formats that the computer can learn efficiently from.
- For example, the speed units can be miles per hour, meters per second, or kilometers per hour. Therefore a dataset may store values of the speed of a car in different units as such.
- Before feeding this data to an algorithm, we need to transform the data into the same unit. Smoothing, Aggregation, Discretization, Generalization, Feature Construction and concept hierarchy generation are some of the techniques used for data transformation.

# Data Cleaning

- The first stage of data preprocessing is Data cleaning which recognizes partial, incorrect, imprecise or inappropriate parts of the data from datasets. Data cleaning may eliminate typographical errors.
- Missing Values The problem of missing data values is quite common. It may happen during data collection or due to some specific data validation rule. In such cases, you need to collect additional data samples or look for additional datasets.

# Data Cleaning

- **Missing Values :-**

- **Ignore the tuple:** This is usually done if the class label is missing (assuming the mining activity requires classification). This solution is not quite good since the tuple contains several attributes with missing values. It is really unfortunate since the percentage of values missed per variable varies greatly.
- **Fill in the missing values manually:** This approach is time intensive and might not be feasible due to a wide range of data with many missing values.
- **Use a global constant to fill in the missing values:** Replace all missing attributes with the same constant, such as a "unknown" or  $\infty$ .



# Data Cleaning

- **Use the attribute mean to fill in the missing value.**
- **Use the mean attribute for all samples belonging to the same class as the given tuple:** When the borrower is identified by credit danger, replace the missing number with the average income value in the same credit risk category as the tuple.
- **Using the most likely meaning to fill in the missing value:** This can be computed using regression, inference-based tools using Bayesian formalism or the introduction of decision tree. For example, in the fixed Decision Tree the use of other market attributes is designed to approximate the missing revenue value.

# Data Cleaning

- **Noisy Data**

- A large amount of meaningless data is called noise. More precisely, it's the random variance in a measured variable or data having incorrect attribute values. Noise includes duplicate or semi-duplicates of data points, data segments of no value for a specific research process, or unwanted information fields.
- Following are the methods for solving the problem of noisy data :-
- **Binning** uses the “neighborhood” to smooth the storage value of the records. That's the value around it. The sorted values are split into “buckets” and “bins”. Since binning methods consult the value community, local smoothing is carried out.

# Data Cleaning

- **Regression:**

- By modifying data to perform, for instance for regression, data may be smoothed. Linear regression specifies that two attributes (or variables) be searched for the “right” line so one attribute can be used to predict the other

- **Clustering:**

- Clustering Outliers may be detected if the same values are clustered in groups or clusters. Values that go outside the spectrum of clusters can intuitively be considered outliers.

# Data Integration

- Data Integration is the method of merging data derived from different sources of data into a consistent dataset. Data on the web is expanding in size and complexity, and is either unstructured or semi-structured. Here are some approaches to integrate data:
  - Data consolidation: Data is physically brought together and stored in a single place. Having all data in one place increases efficiency and productivity.
  - Data virtualization: In this approach, an interface provides a unified and real-time view of data from multiple sources. In other words, data can be viewed from a single point of view.
  - Data propagation: Involves copying data from one location to another with the help of specific applications. This process can be synchronous or asynchronous and is usually event-driven

# Data Integration

- **Data Integration Issues**

- **Schema Integration and Object Matching:** It can be difficult because in different tables, different individuals can identify the same type. This is linked to the problem of individual identity. Metadata can be used to avoid errors in schema integration.
- **Redundancy:** That is another serious issue, that if it can be "derived" from another attribute, an attribute (e.g. annual sales) can be redundant. Inconsistencies in the naming of dimensions can also add to the redundancies in the data collection arising from this.

# Data Integration

- **Detection and Resolution of Data Value Conflicts:** The third major issue in the integration of data is the detection and resolution of conflicts over data. For instance, the significance of attributes from different sources can vary for the same real person – the world person.
- This could be connected to changes in image, size or encoding. Room prices in various cities covering not only different currencies, but also different services (such as free breakfast) and taxes for the hotel chain.

# Data Transformation

- Raw data is usually transformed into a format suitable for analysis. Data can be normalized for instance transformation of the numerical variable to a common range. Categorical data can be transformed using aggregation which merges two or more attributes into a single attribute. Following are some of the strategies for data transformation:
  - **Smoothing** It is a mechanism used to remove data set noise using some algorithms to highlight main data set features. It helps to predict patterns and can be controlled when collecting data to eliminate or reduce any variance or noise.

# Data Transformation

- **Aggregation**

- The compilation or grouping of data is a way of summer data collection and show. In order to integrate these data sources into the concept of data analysis, the data should be obtained from different data sources. This is a vital step because the accuracy of data analysis depends heavily on the volume and quality of the data used.

- **Discretization**

- It is a way to transform continuous data into a number of short intervals. Many actual data mining activities have ongoing attributes. However, many of the existing data mining systems cannot handle these attributes.



# Data Transformation

- **Attribute Construction**

- Where new attributes are created and added to support an attribute set in the mining procedure. This simplifies the original data and improves the efficiency of mining

- **Generalization**

- Convert low - level data attributes with a hierarchical concept to high-level data attributes. For example, an age in numerical form initially (20, 64) is translated into a categorical sense (Young and Old). Categorical attributes such as house addresses, for example, may be applied to higher level definitions such as cities or areas.

# Data Reduction

- The last stage of data preprocessing is data reduction. Multifaceted exploration of huge data sources may consume considerable time or even be infeasible. When the number of predictor variables or the number of instances becomes large, mining algorithms suffer from dimensionality handling problems. Following are some of the strategies for Data Reduction:
  - **Data Cube Aggregation** Data Cube aggregation where a data cube is created using aggregation operations. Data cubes store multidimensional information aggregated. Each cell stores in a multidimensional space a data value corresponding to the data point. For each attribute concept hierarchies may occur, allowing data to be evaluated at multiple abstraction levels.

# Data Reduction

- **Attribute Subset Selection** The selection of a sub-set attribute reduces data by removing redundant attributes or measurements irrelevant. The goal is to find a minimum set of attributes for the selection of a subset attribute. It decreases the number of attributes found in patterns, which promotes comprehension of patterns.
  - Process of Attribute Subset Selection
  - The brute force approach can be very expensive, in which data with  $n$  attributes can be evaluated by any subset ( $2^n$  possible subsets). The simplest way to do this is to use metrics of predictive significance in order to recognise the best or worst attributes.

# Data Reduction

- **Attribute Subset Selection Methods**

- **Stepwise Forward Selection:** The process starts with a minimum number of empty attributes. The most significant attributes with minimal p-value are selected and added to the minimum list. One attribute in each iteration is applied to the reduced collection.
- **Stepwise Backward Elimination** All attributes are considered here in the initial collection of attributes. In each iteration, one attribute is omitted from the set of attributes of p significance.
- **Combination of Forward Selection and Backward Elimination** Phase by step, sorting and reverse exclusion are combined so that the necessary attributes are selected more effectively. This is the most common way to collect attributes

# Data Reduction

- **Decision Tree Introduction** The introduction to the Decision Tree outlines a process flow diagram in which each internal node denotes an attribute test, every branch defines the possible test results and each leaf denotes the class forecast.
- **Dimensionality Reduction** The final definition of machine learning problems is sometimes underlying too many factors. These factors are primarily variables known as features. The higher the number of features, the harder the training set can be imagined and worked on. Most of these features are frequently connected and therefore redundant.

# Data Reduction

- **Significance of Dimensionality Reduction**

- A simple explanation of reduced dimensionality is given by a basic e-mail classification problem, where we have to classify whether mail is spam or not. There is a wide range of features, including whether the e-mail has a general title, e-mail content, whether an e-mail uses a template, etc

- **Components of Dimensionality Reduction**

- **Feature Selection:** Here we try to fine-tune a subset of the original variables or features array to create a smaller subset that can be used for the query. It normally takes three directions namely – Filter, Wrapper and Embedded
- **Feature Extraction:** This limits data to a lower dimension in a high dimensional space, i.e. space with a limited number of dimensions.

# Data Reduction

- **Dimensionality Reduction Methods**

- **Principal component analysis (PCA):** A statistical technique used to extract a new set of variables from a large set of variables. The newly extracted variables are called principal components. This method works only for features with numerical values.
- **High correlation filter:** A technique used to find highly correlated features and remove them; otherwise, a pair of highly correlated variables can increase the multicollinearity in the dataset.
- **Missing values ratio:** This method removes attributes having missing values more than a specified threshold.
- **Low variance filter:** Involves removing normalized attributes having variance less than a threshold value as minor changes in data translate to less information.

# Data Reduction

- **Random forest:** This technique is used to assess the importance of each feature in a dataset, allowing us to keep just the top most important features.
- **Advantages and Disadvantages of Dimensionality Reduction**
  - It helps to compact data and also removes disc space.
  - It reduces the time of calculation
  - It also helps, if any, to eliminate redundant features.
  - This can lead to a certain amount of loss of data.
  - PCA continues to recognize ongoing correlations between variables that are often unwanted



# Data Reduction

- **Numerosity Reduction** :It is a technique for data reduction that replaces the original information with a smaller kind of data representation. There are two ways to minimize numbers.
  - **Parametric Methods** Data is interpreted by any parametric system model. The model is used to simulate data, which means that only data parameters should be processed rather than actual data. To construct these models (i) regression and (ii) log-linear approaches are used.
  - **Non-Parametric Methods** These techniques are used to store reduced data representations such as histograms, clusters, samples, and aggregation of data cubes. Histograms, clustering, sampling and data cube aggregation falls under non-parametric methods.

# Data Reduction

- **Data Discretization and Concept Hierarchy Generation**
  - Discretization techniques can be used to reduce the number of values for a given continuous attribute, by dividing the attribute into a range of intervals. Interval value labels can be used to replace actual data values.
  - Many discretization techniques can be applied recursively in order to provide a hierarchical or multiresolution partitioning of the attribute values known as concept hierarchy.

# Data Reduction

- **Data Discretization Categories**

- **Supervised Discretization:** Uses details regarding the class.
- **Unsupervised Discretization or Splitting:** Does not use class knowledge.
- **Top-Down Discretization or Splitting:** Here, the method begins by first finding one or a few points called break points or cut points to separate the whole set of attributes and then performs this recursively over the subsequent intervals.
- **Bottom – up Discretization or merging:** Here, the method starts by treating all continuous quantities as intervals and then applies this process recursively at the following intervals.

# Data Reduction

- **Discretization and Concept Hierarchy Generation for Numerical data**
  - It is difficult and laborious for to specify concept hierarchies for numeric attributes due to the wide diversity of possible data ranges and the frequent updates if data values. Manual specification also could be arbitrary. Concept hierarchies for numeric attributes can be constructed automatically based on data distribution analysis. Five methods for concept hierarchy generation are defined below-
    - Binning
    - Histogram Analysis
    - Entropy – Based Discretization
    - Interval merging by X2 – Analysis
    - Cluster Analysis

# Important Topics

Purpose of Data Preprocessing

Factors Contributing to Data Quality

Data Cleaning Data Integration

Data Transformation

Data Reduction

Data Cleaning

Data Integration

Data Transformation

Data Reduction

# Summary

- In this session we had studied an important step in data mining i.e., Data Preprocessing. Data pre-processing consists of a series of steps to transform raw data derived from data extraction into a “clean” and “tidy” dataset prior to statistical analysis. We have studied several distinct stages those are involved in preprocessing data such as:-
  - Data Cleaning: This step deals with missing data, noise and duplicate or incorrect records while minimizing introduction of bias into the database.
  - Data Integration: Extracted raw data can come from heterogeneous sources or be in separate datasets. This step reorganizes the various raw datasets into a single dataset that contain all the information required for the desired statistical analyses.

# Summary

- **Data Transformation:** This step translates and/or scales variables stored in a variety of formats or units in the raw data into formats or units that are more useful for the statistical methods that the researcher wants to use.
- **Data Reduction:** After the dataset has been integrated and transformed, this step removes redundant records and variables, as well as reorganizes the data in an efficient and “tidy” manner for analysis.
- **Pre-processing** is sometimes iterative and may involve repeating this series of steps until the data are satisfactorily organized for the purpose of statistical analysis.

*Thank  
You*





# **Data Warehousing and Data Mining**

## **Block-3 Unit-9**

### **Mining Frequent Patterns and Associations**

# Topics to be Covered

- Introduction
- Market Basket Analysis
- Classification of Frequent Pattern Mining
- Association Rule Mining and Related Concepts
- Apriori Algorithm
- Mining Multilevel Association Rules
- Approaches for Mining Multilevel Association Rules
- Mining Multidimensional Association Rules From Relational Databases (RD) And Data Warehouses (DW)
- Mining Quantitative Association Rules
- From Association Mining To Correlation Analysis
- Important Topics
- Summary

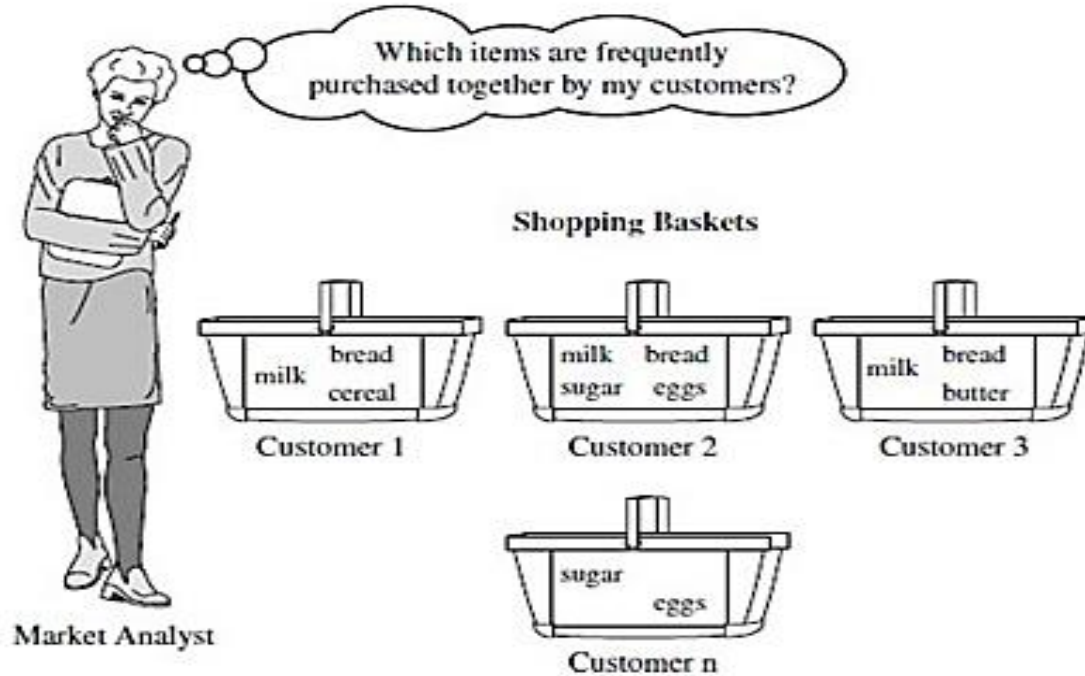
# Introduction

- As we know Data mining technology has emerged as a means for identifying patterns and trends from large quantities of data.
- Data mining, also known as Knowledge Discovery in Databases, has been defined as the nontrivial extraction of implicit, previously unknown, and potentially useful information from data.
- Descriptive mining and Predictive mining are the two categories of data mining tasks.
- The descriptive mining refers to the method in which the essential characteristics or general properties of the data in the database are depicted.

# Market Basket Analysis

- Frequent patterns are patterns (such as itemsets, subsequences, or substructures) that appear in a data set frequently.
- For example, a set of items, such as milk and bread, that appear frequently together in a transaction data set is a frequent itemset.
- Another example may be buying a mobile and its cover appears frequently together in a transaction data set is known as frequent itemset.
- A typical example of frequent itemset mining is market basket analysis.
- This process analyzes customer buying habits by finding associations between the different items that customers place in their “shopping baskets” as shown in Figure below.

# Market Basket Analysis



# Market Basket Analysis

- **Example:** If customers who purchase computers also tend to buy anti-virus software at the same time, then placing the hardware display close to the software display may help increase the sales of both items.
- In an alternative strategy, placing hardware and software at opposite ends of the store may entice customers who purchase such items to pick up other items along the way.
- For instance, after deciding on an expensive computer, a customer may observe security systems for sale while heading toward the software display to purchase antivirus software and may decide to purchase a home security system as well.

# Classification of Frequent Pattern Mining

- Frequent pattern is a pattern which appears frequently in a data set.
- By identifying frequent patterns we can observe strongly correlated items together and easily identify similar characteristics, associations among them.
- By doing frequent pattern mining, it leads to further analysis like clustering, classification and other data mining tasks.
- Frequent pattern mining can be classified in various ways, based on the following criteria:

# Classification of Frequent Pattern Mining

- **Based on the completeness of patterns to be mined:**
  - We can mine the complete set of frequent item sets, the closed frequent item sets, and the maximal frequent item sets, given a minimum support threshold.
  - We can also mine constrained frequent item sets, approximate frequent item sets, near- match frequent item sets, top-k frequent item sets and so on.
- **Based on the levels of abstraction involved in the rule set:**
  - Some methods for association rule mining can find rules at differing levels of abstraction.



# Classification of Frequent Pattern Mining

- **Based on the number of data dimensions involved in the rule:**
  - If the items or attributes in an association rule reference only one dimension, then it is a single-dimensional association rule.
- **Based on the types of values handled in the rule:**
  - If a rule involves associations between the presence or absence of items, it is a Boolean association rule.
  - If a rule describes associations between quantitative items or attributes, then it is a quantitative association rule.

# Classification of Frequent Pattern Mining

- **Based on the kinds of rules to be mined:**
  - Frequent pattern analysis can generate various kinds of rules and other interesting relationships.
  - Association rule mining can generate a large number of rules, many of which are redundant or do not indicate a correlation relationship among item sets.
- **Based on the kinds of patterns to be mined:**
  - Many kinds of frequent patterns can be mined from different kinds of data sets.
  - Sequential pattern mining searches for frequent subsequences in a sequence data set, where a sequence records an ordering of events.

# Association Rule Mining and Related Concepts

- An association rule consists of a set of items, the rule body, leading to another item, the rule head. The association rule relates the rule body with the rule head. An association rule can contain the following characteristics:
  - Statistical information about the frequency of occurrence
  - Reliability
  - Importance of this relation

# Association Rule Mining and Related Concepts

- **Association Rule Mining**

- One of the popular descriptive data mining techniques is Association rule mining (ARM), owing to its extensive use in marketing and retail communities in addition to many other diverse fields.
- Mining association rules is particularly useful for discovering relationships among items from large databases.

# Association Rule Mining and Related Concepts

- **Some Important Concepts :-**

- **Itemset** A set of items together is called an itemset. For Example, Bread and butter, Laptop and Antivirus software, etc are itemsets.
- **k-Itemset** An itemset is just a collection or set of items. k-itemset is a collection of k items. For e.g. 2-itemset can be {egg,milk} or {milk,bread} etc. Similarly 3-itemset example can be {egg,milk,bread} etc.
- **Support Count** Support count is the number of times the itemsets appears out of all the transactions under consideration. It can also be expressed in percentage. A support count of 65% for {milk, bread} means that out both milk and bread appeared 65 times in the overall transaction of 100.

# Association Rule Mining and Related Concepts

- **Support** A transaction supports an association rule if the transaction contains the rule body and the rule head. The rule support is the ratio of transactions supporting the association rule and the total number of transactions within your database of transactions.
- **Confidence** The confidence of an association rule is its strength or reliability. The confidence is defined as the percentage of transactions supporting the rule out of all transactions supporting the rule body. A transaction supports the rule body if it contains all the items of the rule body.

# Association Rule Mining and Related Concepts

- **Applications of Association Rule Mining** : Association rule mining is used in different areas, as it is very advantageous. Some of the fields that have adopted association rule mining are discussed below:-
  - **Market Basket Analysis**: One of the best and typical examples of association rule mining is market basket analysis, for example in supermarket stores. Managers of these stores want to increase the interest of the existing customers and to attract the new ones. Such stores consist of large no of databases and there is huge number of transactional records. So the managers may be interested to know whether some items are consistently purchased.

# Association Rule Mining and Related Concepts

- **CRM of the Credit Card:** Business Customer Relationship Management (CRM) is a system that is used by a bank to manage its relationships with the customers. Banks usually identify the behavior of their customers to find out their likings and interests. In this way banks can increase the coherence of credit card customers and the bank.
- **Medical Diagnosis:** Association rules are also used in the field of medical sciences. It is helpful for the physicians to cure the patients. Association rules are used to find out the probability of illness in a disease. There is a need for proper explanation of a disease. Thus diagnosis in itself is not an easy task. Adding some new symptoms to an existing disease, and then finding out the relationships between the symptoms will help the physicians



# Association Rule Mining and Related Concepts

- **Census Data:** Association rule mining has a great possibility in census data. Association rules helps in supporting good public policy and in business development.
- Huge statistical information is generated by census. The information is related to the economic and population census and thus can be used in planning public services and business.
- In services, it includes health, education, transport etc. In business, it has constructing new malls, factories and banks.

# Apriori Algorithm

- An algorithm known as Apriori is a common one in data mining. It's used to identify the most frequently occurring elements and meaningful associations in a dataset.
  - The name of the algorithm is based on the fact that the algorithm uses prior knowledge of frequent itemset properties.
  - Apriori employs an iterative approach known as a level-wise search, where  $k$ -item sets are used to explore  $(k+1)$ -itemsets.

# Apriori Algorithm

- First, the set of frequent 1-itemsets is found by scanning the database to accumulate the count for each item, and collecting those items that satisfy minimum support.
- The resulting set is denoted L1. Next, L1 is used to find L2, the set of frequent 2-itemsets, which is used to find L3, and so on, until no more frequent k-item sets can be found.
- The finding of each  $L_k$  requires one full scan of the database.
- A two-step process is followed in Apriori consisting of join and prune action.

# Mining Multilevel association rules

- Multilevel association rules can be mined efficiently using concept hierarchies under a support-confidence framework.
- In general, a top-down strategy is employed, where counts are accumulated for the calculation of frequent item sets at each concept level, starting at the concept level 1 and working downward in the hierarchy toward the more specific concept levels, until no more frequent item sets can be found.
- The concept hierarchy has different levels :-

# Mining Multilevel association rules

Here, Level 1 includes computer, software, printer & camera, and computer accessory.

Level 2 includes laptop computer, desktop computer, office software, antivirus software

Level 3 includes IBM desktop computer. . . Microsoft office software and so on.

Level 4 is the most specific abstraction level of this hierarchy.

# Approaches for Mining Multilevel association rules

- **Uniform Minimum Support**

- The same minimum support threshold is used when mining at each level of abstraction.
- When a uniform minimum support threshold is used, the search procedure is simplified.
- The method is also simple in that users are required to specify only one minimum support threshold.
- The uniform support approach, however, has some difficulties. It is unlikely that items at lower levels of abstraction will occur as frequently as those at higher levels of abstraction.

# Approaches for Mining Multilevel association rules

- **Reduced Minimum Support**

- Each level of abstraction has its own minimum support threshold.
- The deeper the level of abstraction, the smaller the corresponding threshold is.
- For example, the minimum support thresholds for levels 1 and 2 are 5% and 3%, respectively. In this way, computer, laptop computer, and desktop computer are all considered frequent.

# Approaches for Mining Multilevel association rules

- **Group-Based Minimum Support**

- Because users or experts often have insight as to which groups are more important than others, it is sometimes more desirable to set up user-specific, item, or group based minimal support thresholds when mining multilevel rules.
- For example, a user could set up the minimum support thresholds based on product price, or on items of interest, such as by setting particularly low support thresholds for laptop computers and flash drives in order to pay particular attention to the association patterns containing items in these categories.



# Mining Multidimensional association rules from RD And DW

- Single dimensional or intra dimensional association rule contains a single distinct predicate (e.g., buys) with multiple occurrences i.e., the predicate occurs more than once within the rule.
- $\text{buys}(X, \text{—digital camera}\|) \Rightarrow \text{buys}(X, \text{—HP printer}\|)$
- Association rules that involve two or more dimensions or predicates can be referred to as multidimensional association rules.

# Mining Multidimensional association rules from RD And DW

- Multidimensional association rules with no repeated predicates are called inter dimensional association rules.
- We can also mine multidimensional association rules with repeated predicates, which contain multiple occurrences of some predicates.
- These rules are called hybrid- dimensional association rules. An example of such a rule is the following, where the predicate buys is repeated:
- $\text{age}(X, \text{—}20\dots29\|) \wedge \text{buys}(X, \text{—laptop}\|) \Rightarrow \text{buys}(X, \text{—HP printer}\|)$

# Mining Quantitative association rules

Quantitative association rules are multidimensional association rules in which the numeric attributes are dynamically discretized during the mining process so as to satisfy some mining criteria, such as maximizing the confidence or compactness of the rules mined.

# From Association Mining to correlation analysis

- A correlation measure can be used to augment the support-confidence framework for association rules. This leads to correlation rules of the form :

$A \Rightarrow B$  [support, confidence, correlation]

- That is, a correlation rule is measured not only by its support and confidence but also by the correlation between item sets A and B. There are many different correlation measures from which to choose.

# Important Topics

Apriori Algorithm

Mining Multilevel Association Rules

Approaches for Mining Multilevel Association Rules

Mining Multidimensional Association Rules From Relational  
Databases And Data Warehouses

# Summary

- In this session we had studied the concepts like itemset, frequent itemset, market basket analysis, frequent itemset mining, association rules, association rule mining, Apriori algorithm along with some advanced concepts.
- **Data Transformation:** This step translates and/or scales variables stored in a variety of formats or units in the raw data into formats or units that are more useful for the statistical methods that the researcher wants to use.

# Summary

- **Data Reduction:** After the dataset has been integrated and transformed, this step removes redundant records and variables, as well as reorganizes the data in an efficient and “tidy” manner for analysis.
- Pre-processing is sometimes iterative and may involve repeating this series of steps until the data are satisfactorily organized for the purpose of statistical analysis.

*Thank  
You*





# **Data Warehousing and Data Mining**

## **Block-4 Unit-10 Part-2**

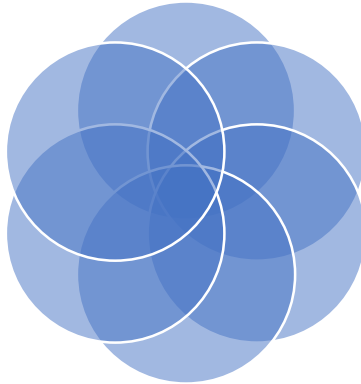
### **Classification**

# Topics to be Covered

Introduction

Summary

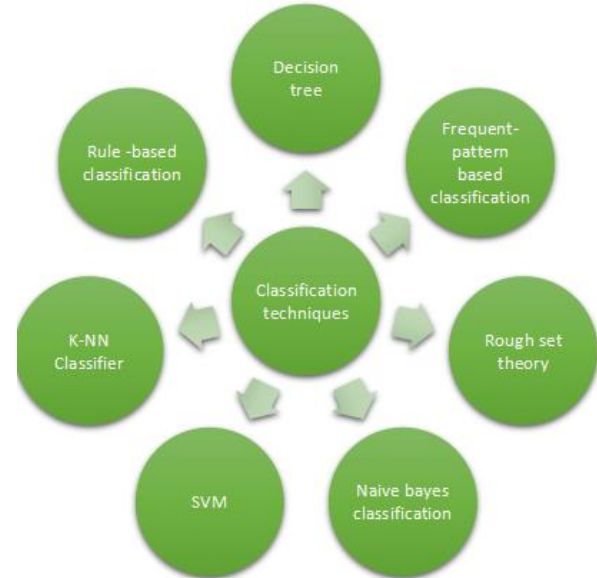
Important  
Topics



Model  
Evaluation  
and Selection

Support  
Vector  
Machines

Rule Based  
Classification



<https://www.geeksforgeeks.org/classification-based-approaches-in-data-mining/>

# Introduction

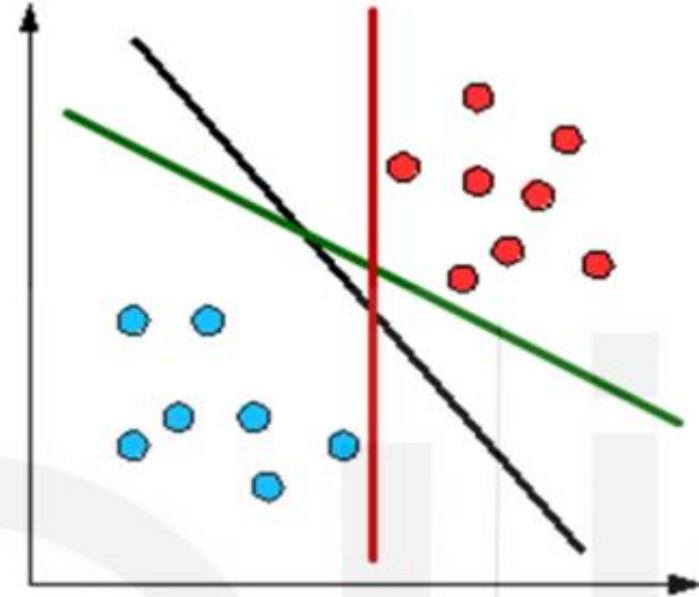
- KDD is one of the most commonly used technique's when it comes to classifying large sets of data.
- This method of data analysis includes algorithms adapted to the data quality. The algorithm that performs the classification is the classifier while the observations are the instances
- In this session we are going to study the concept of classification in data mining and various classifiers.

# Support Vector Machines

- **SVM**
- It is a supervised learning algorithm which can be used to solve both classification and regression problem, even though the current focus is on classification only.
- This algorithm looks for a linearly separable hyperplane, or a decision boundary separating members of one class from the other. If such a hyperplane exists, the work is done! If such a hyperplane does not exist, SVM uses a nonlinear mapping to transform the training data into a higher dimension.

# Support Vector Machines

- **When Data is Linearly Separable**
  - Let the  $i$ -th data point be represented by  $(X_i, y_i)$  where  $X_i$  represents the feature vector and  $y_i$  is the associated class label, taking two possible values  $+1$  or  $-1$ . In the Figure 2, above the balls having red color has class label  $+1$  and the blue balls have class label  $-1$ , say. A straight line can be drawn to separate all the members belonging to class  $+1$  from all the members belonging to the class  $-1$ . The two dimensional data above are clearly linearly separable.



# Support Vector Machines

In fact, an infinite number of straight lines can be drawn to separate the blue balls from the red balls.

The problem therefore is which among the infinite straight lines is optimal, in the sense that it is expected to have minimum classification error on a new observation.

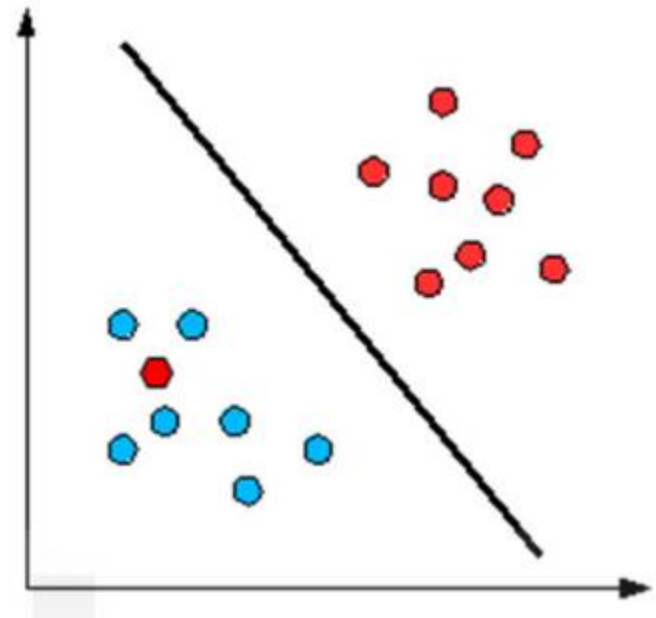
The straight line is based on the training sample and is expected to classify one or more test samples correctly.

# Support Vector Machines

- The maximal margin classifier is a very natural way to perform classification, is a separating hyperplane exists.
- However existence of such a hyperplane may not be guaranteed, or even if it exists, the data is noisy so that maximal margin classifier provides a poor solution.
- In such cases, the concept can be extended where a hyperplane exists which almost separates the classes, using what is known as a soft margin.

# Support Vector Machines

- **When Data is Not Linearly Separable**
- SVM is quite intuitive when the data is linearly separable.
- However, when they are not, as shown in the Figure, SVM can be extended to perform well.





# Support Vector Machines

- There are two main steps for nonlinear generalization of SVM. The first step involves transformation of the original training (input) data into a higher dimensional data using a nonlinear mapping.
- Once the data is transformed into the new higher dimension, the second step involves finding a linear separating hyperplane in the new space.
- The maximal marginal hyperplane found in the new space corresponds to a nonlinear separating hyper-surface in the original space.

# Rule Based Classification

- Rule-based classifier makes use of a set of IF-THEN rules for classification. We can express a rule in the following form :-
- **IF condition THEN conclusion**
- Let us consider a rule R1,
- R1: IF age = youth AND student = yes
- THEN buy\_computer = yes
- Rule-Based Classifier classifies records by using a collection of “if...then...” rules.

# Rule Based Classification

- **Example 1**

- $(\text{BloodType} = \text{Warm}) \wedge (\text{LayEggs} = \text{Yes}) \rightarrow \text{Birds}$
- $(\text{TaxableIncome} < 50\text{K}) \vee (\text{Refund} = \text{Yes}) \rightarrow \text{Evade} = \text{No}$

- **Example 2**

| Name   | Blood Type | Give Birth | Can Fly | Live in Water | Class      |
|--------|------------|------------|---------|---------------|------------|
| human  | warm       | yes        | no      | no            | mammals    |
| python | cold       | no         | no      | no            | reptiles   |
| salmon | cold       | no         | no      | yes           | fishes     |
| whale  | warm       | yes        | no      | yes           | mammals    |
| frog   | cold       | no         | no      | sometimes     | amphibians |
| komodo | cold       | no         | no      | no            | reptiles   |
| bat    | warm       | yes        | yes     | no            | mammals    |

# Rule Based Classification

## Advantages of Rule-Based Classifiers

- They are as highly expressive as decision trees -Easy to interpret
- They are easy to generate
- They can classify new instances rapidly
- Their performance is comparable to decision trees

# Rule Based Classification

- Coverage and Accuracy Example

| <i>Tid</i> | <b>Refund</b> | <b>Marital Status</b> | <b>Taxable Income</b> | <b>Class</b> |
|------------|---------------|-----------------------|-----------------------|--------------|
| 1          | Yes           | Single                | 125K                  | No           |
| 2          | No            | Married               | 100K                  | No           |
| 3          | No            | Single                | 70K                   | No           |
| 4          | Yes           | Married               | 120K                  | No           |
| 5          | No            | Divorced              | 95K                   | Yes          |
| 6          | No            | Married               | 60K                   | No           |

# Rule Based Classification

- **Coverage of a rule:**
  - Fraction of all records that satisfy the antecedent of a rule
  - $\text{Count}(\text{instances with antecedent}) / \text{Count}(\text{training set})$
  - Example on left: (Status = 'Single')  $\rightarrow$  no, Coverage =  $4/10 = 40\%$
- **Accuracy of a rule:**
  - Fraction of records that satisfy the antecedent that also satisfy the consequent of a rule
  - $\text{Count}(\text{instances with antecedent AND consequent}) / \text{Count}(\text{instances with antecedent})$
  - Example on left: (Status = 'Single')  $\rightarrow$  no, accuracy =  $2/4 = 50\%$

# Rule Based Classification

- **Characteristics of Rule Sets** : If we convert the result of decision tree to classification rules, these rules would be mutually exclusive and exhaustive at the same time.
- **Mutually Exclusive rules** :
  - Classifier contains mutually exclusive rules if the rules are independent of each other.
  - Every record is covered by at most one rule
- **Exhaustive rules**
  - Classifier has exhaustive coverage if it accounts for every possible combination of attribute values.
  - Each record is covered by at least one rule.

# Rule Based Classification

**Ordered Rule Set :** An ordered rule set is known as a decision list. Rules are rank ordered according to their priority.

For example, when a test record is presented to the classifier, it is assigned to the class label of the highest ranked rule it has triggered. If none of the rules fired, it is assigned to the default class.



# Rule Based Classification

- **Rule Ordering Schemes** : If more than one rule is triggered, it needs conflict resolution in the following ways:
  - Size ordering - assign the highest priority to the triggering rules that has the “toughest” requirement (i.e., with the most attribute test)
  - Class-based ordering - decreasing order of prevalence or misclassification cost per class
  - Rule-based ordering (decision list) - rules are organized into one long priority list, according to some measure of rule quality or by experts

# Rule Based Classification

## Advantages of Rule- Based Classifiers :

- They have the characteristics quite similar to decision trees
- These classifiers are as highly expressive as decision trees
- They are easy to interpret
- Their performance is comparable to decision trees
- They can handle redundant attributes
- They are better suited for handling imbalanced classes
- They are harder to handle missing values in the test set

# Model Evaluation and Selection

- Evaluating the performance of a data mining technique is a fundamental aspect of data mining. Evaluation method is the yardstick to examine the efficiency and performance of any model.
- The evaluation is important for understanding the quality of the model or technique, for refining parameters in the iterative process of learning.
- **Model selection** is a technique for selecting the best model after the individual models are evaluated based on the required criteria. Model selection is the problem of choosing one from among a set of candidate models. It is common to choose a model that performs the best on a hold-out test dataset or to estimate model performance using a resampling technique, such as k-fold cross-validation.

# Model Evaluation and Selection

- **Types of Model Selection :**

- The simplest reliable method of model selection involves fitting candidate models on a training set, tuning them on the validation dataset, and selecting a model that performs the best on the test dataset according to a chosen metric, such as accuracy or error. A problem with this approach is that it requires a lot of data
- Resampling techniques attempt to achieve the same as the train/val/test approach to model selection, although using a small dataset.
- A third approach to model selection attempts to combine the complexity of the model with the performance of the model into a score, then select the model that minimizes or maximizes the score.

# Model Evaluation and Selection

- **Models Evaluation** : Models can be evaluated using multiple metrics. However, the right choice of an evaluation metric is crucial and often depends upon the problem that is being solved. A clear understanding of a wide range of metrics can help the evaluator to choose upon an appropriate match of the problem statement and a metric.
- **Classification metrics** For every classification model prediction, a matrix called the confusion matrix can be constructed which demonstrates the number of test cases correctly and incorrectly classified.
- **Confusion Matrix** A binary classification model classifies each instance into one of two classes; say a true and a false class.

# Model Evaluation and Selection

- Precision : Precision is the metric used to identify the correctness of classification.
- Recall : Recall tells us the number of positive cases correctly identified out of the total number of positive cases.
- F1- Score : is also known as F-Score is the harmonic mean of Recall and Precision and therefore, balances out the strengths of each.
- Receiver Operating Curves (ROC) : Central to constructing, deploying, and using classification models is the question of model performance assessment. Traditionally this is accomplished by using metrics derived from the confusion matrix or contingency table.

# Important Topics

Classification in Data Mining

Decision Tree Classifier

Support Vector Machines



# Summary

- In this session we have seen that support Vector Machine: Support Vector Machines (SVM), is a classification technique that can be used to build both classifiers and non-parametric regression models. SVM works by finding an optimal hyperplane that separates objects of different classes in the input space based on their training samples.
- A hyper plane with a good non-linear mapping to a high enough dimension can always divide data into two groups.
- SVM uses support vectors (“important training tuples”) and margins (specified by the support vectors) to discover this hyper plane. SVM is used for classification as well as prediction.



*Thank  
You*



# **Data Warehousing and Data Mining**

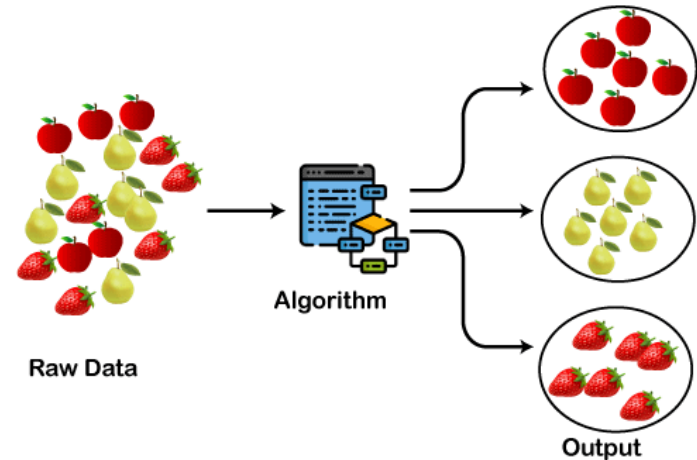
## **Block-4 Unit-11**

### **Clustering**

# Topics to be covered

- Introduction
- Clustering – An Overview
- Clustering Methods
- Partitioning Method
- Hierarchical Method
- Density Based Method
- Limitations with Cluster Analysis
- Outlier Analysis

- Important Topics
- Summary



<https://www.javatpoint.com/clustering-in-machine-learning>

# Introduction

- This session covers another important concept known as Clustering.
- Clustering is the process by which we create groups in a data, like customers, products, employees, text documents, in such a way that objects falling into one group exhibit many similar properties with each other and are different from objects that fall in the other groups that got created during the process.
- We will study overview of clustering, clustering methods, partitioning method, hierarchical clustering and outlier analysis.

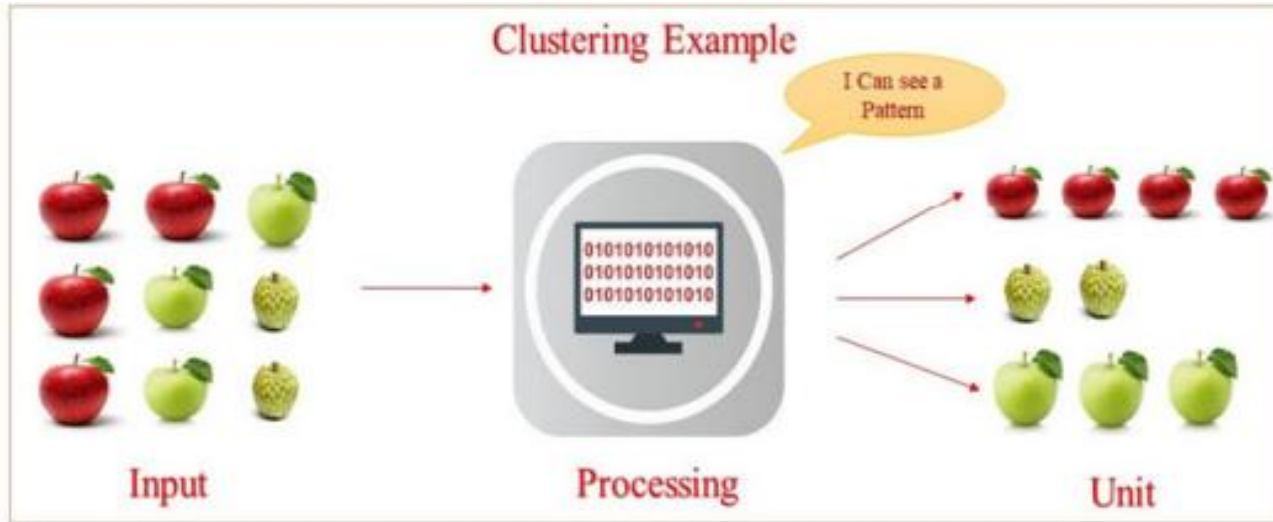
# Clustering – An Overview

- Clustering is the process of grouping a collection of objects (usually represented as points in a multidimensional space) into classes of similar objects. Cluster analysis is a very important tool in data analysis. It is a set of methodologies for automatic classification of a collection of patterns into clusters based on similarity.
- **Typical pattern clustering activity involves the following steps:**
  - pattern representation (including feature extraction and/or selection)
  - definition of a pattern proximity measure appropriate to the data domain
  - Clustering
  - data abstraction

# Clustering – An Overview

- Clustering is an unsupervised-based algorithm that divides a set of data points into clusters so that the objects are all part of the same group. Using clustering, data can be subdivided into smaller, more manageable groups. Data from each of these subgroups are grouped together into a single cluster.
  - To put it simply, a cluster is a group of related things.
  - It's a grouping of objects where the distance between any two members is less than the distance between any two people.
  - A multidimensional space segment with a high density of objects that is related to the other segments.
  - A simple example of Clustering is illustrated in Figure as shown below:

# Clustering – An Overview



- Clustering helps in organizing huge voluminous data into clusters and displays interior structure of statistical information.

# Clustering – An Overview

- **Applications of Cluster Analysis in Data Mining**

- It aids data discovery by assigning documents on the Internet.
- Credit card fraud detection relies on clustering.
- Cluster analysis is a data mining function that provides insight into data distribution so that the properties of each cluster may be analyzed.
- It can be used in biology to figure out plant and animal taxonomies, classify genes with similar functions, and have a better understanding of population structure.
- OTT platforms are using clustering in implementing movie recommendations for its users



# Clustering Methods

- **For a successful grouping there are two major goals :**
  - similarity between one data point with another
  - distinction of those similar data points with others which most certainly, heuristically differ from those points.
- The basis of such divisions begins with our ability to scale large datasets and that's a major beginning point for us. This data contains different kinds of attributes like categorical data, continuous data etc.. Dealing with these is the second challenge. The next challenge is the multidimensional data. The clustering algorithm should successfully cross this hurdle as well.

# Clustering Methods

- **The following are various types of Clustering methods:**
  - **Partitioning Method** Partitioning methods breaks down the data into a set of different clusters. Given  $n$  objects, this method produces  $k$  clusters of data where  $k < n$  clusters of data and using an iterative relocation method. Algorithms used in partitioned clustering are k-means algorithm and k-medoids algorithm.
  - **Partitioning Method** Partitioning methods breaks down the data into a set of different clusters. Given  $n$  objects, this method produces  $k$  clusters of data where  $k < n$  clusters of data and using an iterative relocation method. Algorithms used in partitioned clustering are k-means algorithm and k-medoids algorithm.

# Clustering Methods

- **Density-Based Method** Density based clustering is clustering of database based on the densities (i.e. local cluster criterion). It has major features such as:
  - It discovers the arbitrary shape cluster.
  - It handles the noise data.
  - It examines only the local region to justify the density.
  - To termination the process it requires the density parameters.

# Clustering Methods

- **Grid Based Method** The idea of grid-based clustering methods is based on the clustering oriented query answering in multilevel grid structures.
- The upper level stores the summary of the information of its next level, thus the grids make cells between the connected levels.
- Many grid-based methods have been proposed, such as STING (Statistical Information Grid Approach) CLIQUE, and the combination of grid-density based technique.
- The grid-based methods are efficient on clustering data with the complexity of  $O(N)$ . However the primary issue of grid-based techniques is how to decide the size of grids. This depends on the user's experience.

# Clustering Methods

- **Model Based Clustering** Method Model-based clustering methods are based on the assumption that data are generated by a mixture of underlying probability distributions, and they optimize the fit between the data and some mathematical model, for example statistical approach, neural network approach and other AI approaches.
- When facing an unknown data distribution, choosing a suitable one from the model based candidates is still a major challenge. On the other hand, clustering based on probability suffers from high computational cost, especially when the scale of data is very large.

# Clustering Methods

- **Constraint Based Method** The clustering process, in general, is based on the approach that the data can be divided into an optimal number of “unknown” groups.
- The underlying stage of all the clustering algorithms is to find those hidden patterns and similarities, without any intervention or predefined conditions.
- However, in certain business scenarios, we might be required to partition the data based on certain constraints.

# Partitioning Method

- This method is one of the most popular choices for analysts to create clusters.
- This is also known as Supervised Clustering method. In partitioning clustering, the clusters are partitioned based upon the characteristics of the data points.
- We need to specify the number of clusters to be created for this clustering method.
- The algorithms that fall into this category are as follows:

# Partitioning Method

- **k-Means Algorithm**

- The main aim of this algorithm is to find out the location of the clusters thus minimizing the distance between the cluster and the data set.

- **Algorithm**

- Define the number of clusters ( $k$ ) to be produced and identical data point centroids.
- The distance from every data point to all the centroids is calculated and the point is assigned to the cluster with a minimum distance.
- Follow the above step for all the data points.
- The average of the data points present in a cluster is calculated and can set new centroid for that cluster.
- Until desired clusters are formed repeat Step 2.



# Partitioning Method

- **Advantages**

- Effortless implementation process.
- Dense clusters are produced when clusters are spherical when compared to hierarchical method.
- Appropriate for large databases.

- **Disadvantages**

- Inappropriate for clusters with different density and size.
- Equivalent results are not produced on iterative run.
- Euclidean distance measures can weigh unequally due to underlying factors.
- Unsuccessful for non-linear data set and categorical data.

# Partitioning Method

- **k-Medoids or PAM (Partitioning Around Medoids)** - It is similar in process to the K-means clustering algorithm with the difference being in the assignment of the center of the cluster. In PAM, the medoid of the cluster has to be an input data point while this is not true for K-means clustering as the average of all the data points in a cluster may not belong to an input data point.
- The algorithm is implemented in two steps:
  - **Build:** Initial medoids are innermost objects.
  - **Swap:** A function can be swapped by another function until the function can no longer be reduced.

# Partitioning Method

- **Algorithm**

- 1. Initially choose  $m$  random points as initial medoids from given data set.
- 2. For every data point assign a closest medoid by distance metrics.
- 3. Swapping cost is calculated for every chosen and unchosen object given as TCns where  $s$  is selected and  $n$  is non-selected object.
- 4. If  $TCns < 0$ ,  $s$  is replaced by  $n$
- 5. Until there is no change in medoids, repeat 2 and 3.

# Partitioning Method

- **Four characteristics to be considered are:**
  - **Shift-out membership:** Movement of an object from current cluster to another is allowed.
  - **Shift-in membership:** Movement of an object from outside to current cluster is allowed.
  - **Update the current medoids:** Current medoid can be replaced by a new medoid.
  - **No change:** Objects are at their appropriate distances from cluster.

# Partitioning Method

- **Advantages**

- Effortless understanding and implementation process.
- Can run quickly and converge in few steps.
- Dissimilarities between the objects is allowed.
- Less sensitive to outliers when compared to k-means.

- **Disadvantages**

- Initial sets of medoids can produce different clustering's. It is thus advisable to run the procedure several times with different initial sets.
- Resulting clusters may depend upon units of measurement. Variables of different magnitude can be standardized.

# Hierarchical Method

- This method decomposes a set of data items into a hierarchy. Depending on how the hierarchical breakdown is generated, we can put hierarchical approaches into different categories. Following are the two approaches;
  - **Agglomerative Approach** : This Algorithm is also referred as Bottom-up approach. This approach treats each and every data point as a single cluster and then merges each cluster by considering the similarity (distance) in each individual cluster until a single large cluster is obtained or when some condition is satisfied.

# Hierarchical Method

- **Algorithm**

- Initialize all  $n$  data points into  $N$  individual clusters.
- Find the cluster pairs with the least distance (closest distance) and combine them as one single cluster.
- Calculate pair-wise distance between the clusters at present that is the new formed cluster and the priority available clusters.
- Repeat steps 2 and 3 until all data samples are merged into a single large cluster of size  $N$

# Hierarchical Method

- **Advantages**

- Easy to identify nested clusters.
- Gives better results and ease in implementation.
- They are suitable for automation.
- Reduces the effect of initial values of cluster on the clustering results.
- Reduces the computing time and space complexity.

- **Disadvantages**

- It can never undo what was done previously.
- Difficulty in handling different sized clusters and convex shapes lead to increase in time complexity



# Hierarchical Method

- **Divisive Approach** : This approach is also referred as the top-down approach. In this, we consider the entire data sample set as one cluster and continuously splitting the cluster into smaller clusters iteratively. It is done until each object in one cluster or the termination condition holds. This method is rigid, because once a merging or splitting is done, it can never be undone.
- **Algorithm**
  - Initially, initiate the process with one cluster containing all the samples.
  - Select a largest cluster from the cluster that contains widest diameter.

# Hierarchical Method

- Detect the data point in the cluster found in step 2 with the minimum average similarity to the other elements in that cluster.
- The first element to be added to the fragment group is the data samples found in step 3. 5. Detect the element in the original group which has the highest average similarity with the fragment group.
- If the average similarity of element obtained in step 5 with the fragment group is greater than its average similarity with the original group then assign the data sample to the fragment group and go to step 5; otherwise do nothing;
- Repeat the step 2 to 6 until each data point is separated into individual cluste

# Hierarchical Method

- **Advantage**

- It produces more accurate hierarchies than bottom-up algorithm in some circumstances.

- **Disadvantages**

- Top down approach is computationally more complex than bottom up approach because we need a second flat clustering algorithm.
- Use of different distance metrics for measuring distance between clusters may generate different results.

# Density Based Method

- The primary idea of density-based methods is that for each point of a cluster the neighborhood of a given unit distance contains at least a minimum number of points, i.e. the density in the neighborhood should reach some threshold. However, this idea is based on the assumption of that the clusters are in the spherical or regular shapes.
  - **DBSCAN(Density-Based Spatial Clustering of Applications with Noise)**  
DBSCAN was proposed to adopt density-reachability and density connectivity for handling the arbitrarily shaped clusters and noise. In DBSCAN, a cluster is defined as group of data that is of highly dense.

# Density Based Method

- **Algorithm**

- In order to form clusters, initially consider a random point say point  $p$
- The second step is to find the all points that are density reachable from point  $p$  with respect to  $Eps$  and  $MinPts$ . The following condition is checked in order to form the cluster
  - If point  $p$  is found to be core point, then cluster is obtained.
  - If point  $p$  is found to be border point, then no points are density reachable from point  $p$  and hence visit the next point of database.
- Continue this process until all the points is processed.

# Density Based Method

- **Advantages**

- It can identify Outlier.
- It does not require number of clusters to be specified in advance.

- **Disadvantages**

- If the density of data keeps changing then efficiency of finding clusters is difficult.
- It does not suit for high quality of data and the user has to specify the parameter in advance.

# Limitations with Cluster Analysis

- The existing automated clustering algorithms on dealing with arbitrarily shaped data distribution of the datasets.
- The second issue is that, the evaluation of the quality of clustering results by statistics-based methods is time consuming when the database is large, primarily due to the drawback of very high computational cost of statistics based methods for assessing the consistency of cluster structure between the sampling subsets.

# Limitations with Cluster Analysis

- In addition, the inefficiency of clustering algorithms on handling arbitrarily shaped clusters in extremely large datasets directly impacts the effect of cluster validation, because cluster validation is based on the analysis of clustering results produced by clustering algorithms.
- Moreover, most of the existing clustering algorithms tend to deal with the entire clustering process automatically, i.e., once the user sets the parameters of algorithms, the clustering result is produced with no interruption, which excludes the user until the end.



# Outlier Analysis

- In statistics, an outlier is a data point that deviates significantly from the norm. Measuring and execution errors are two possible causes. Outlier analysis, also known as outlier mining, is the process of examining data that stands out for whatever reason. It's impossible to do data analysis without encountering outliers.
- Outliers, on the other hand, are distinct from random noise in the following way:
  - An outlier is a point of observation that differs from the rest because of its location.
  - To better detect outliers, noise should be eliminated

# Outlier Analysis

- **Outliers in Data Mining** For the most part, data mining algorithms ignore outliers like noise and exceptions. However, in certain applications like fraud detection, uncommon occurrences can be just as fascinating as the more common ones, therefore performing an outlier analysis becomes critical. Outliers in Data Mining can have a variety of causes. Following are a few of these contributing factors:
  - Identifying financial fraud such as credit card hacking or other similar scams makes use of this technology.
  - It's utilized to keep track of a customer's changing purchase habits.
  - It's used to find and report human-made mistakes in typing.

# Outlier Analysis

- **Handling of Outliers in Data Mining**

- Data Mining must deal with outliers for a variety of reasons. These are a few of the explanations:
  - The outcomes of databases are impacted by outliers.
  - Outliers frequently produce good or useful discoveries and conclusions, allowing researchers to identify various patterns or trends.
  - Even in the world of study, outliers can be useful. They can be a lifesaver when doing research.
  - Data mining's most important subfield is outlier analysis.

# Outlier Analysis

- **Outlier Detection**

- Outlier detection can be defined as the process of detecting and then excluding outsiders from a given set of data.
- There are no standardized outlier identification methods because these are mostly dataset-dependent.
- Outlier detection as a branch of data processing has many applications in data stream analysis.
- There are four Outlier Detection techniques in general.

# Outlier Analysis

- **Numeric Outlier:** A numerical outlier is a simple, non-standard outlier detection technique in a one dimensional feature space. Extremes are calculated by IQR (Inter Quartile Range). For example, the first and third quarters (Q1, Q3) are calculated. Outlier is a data point  $x_i$  that is out of range.
- **Z-Score:** The Z-score technique considers the Gaussian distribution of data. Outliers are data points that are on the tail of the distribution and are therefore far from average.
- **DBSCAN:** This outlier detection technique is based on the DBSCAN clustering method. DBSCAN is a non-standard, density-based outlier detection method. Here, all data points are defined as focal points, boundary points, or noise points.

# Outlier Analysis

- **Isolated forest** This non-parameter system is suitable for large datasets in one or more dimensional features. Isolation number is very important in this outlier detection technique. Isolation number is the number of divisions required to isolate a data point.
- **Models for Outlier Detection** Analysis There are many approaches to detecting abnormalities. Outlier detection models can be classified into the following groups:
  - **Intensive Value Analysis** Extreme value analysis is the most basic form of outlier detection and is suitable for 1- dimensional data. In this external analysis approach, the largest or smallest values are considered externally.

# Outlier Analysis

- **Linear Models** In this approach, data is structured outside the lower dimensional substructure using linear interactions. The distance of each data point is calculated for a plane that corresponds to the sub-interval. This distance is used to detect outliers. PCA (primary component analysis) is an example of a linear model for anomaly detection.
- **Probabilistic and Statistical Models** In this approach, probability and statistical models consider specific distributions of data. Expectation-enhancement (EM) methods are used to estimate the parameters of the sample.

# Outlier Analysis

- **Proximity-based Models** In this mode, the outliers are designed as points of isolation from other observations. Cluster analysis, density-based analysis, and neighborhood environment are key approaches of this type.
- **Information-theoretical models** In this mode, outliers increase the minimum code length to describe a data set.
- **Uses for Detecting Outliers in Data Mining** In Data Mining, it is common to utilize Outlier Detection to find anomalies. In data mining, it's utilized to find patterns or trends. The following are some examples of where outlier detection is put to use in data mining:



# Outlier Analysis

Fraud Detection

Telecom Fraud  
Detection

Intrusion Detection  
in Cyber Security

Medical Analysis

Environment  
Monitoring such as  
Cyclone, Tsunami,  
Floods, Drought  
and so on

Noticing  
unforeseen entries  
in Databases

# Important Topics

Clustering

Clustering Methods

Partitioning Method

Hierarchical Method

Density Based Method

# Summary

- In this session we had studied the introductory topics of clustering, clustering methods, algorithms associated with clustering and outlier analysis.
- Cluster Analysis refers to the process of identifying groupings of items that are similar in some respects but differ from one another in other respects. It is a grouping of comparable items that belongs to the same category.
- Cluster Analysis refers to the process of identifying groupings of items that are similar in some respects but differ from one another in other respects.
- It is a grouping of comparable items that belongs to the same category.

*Thank  
You*



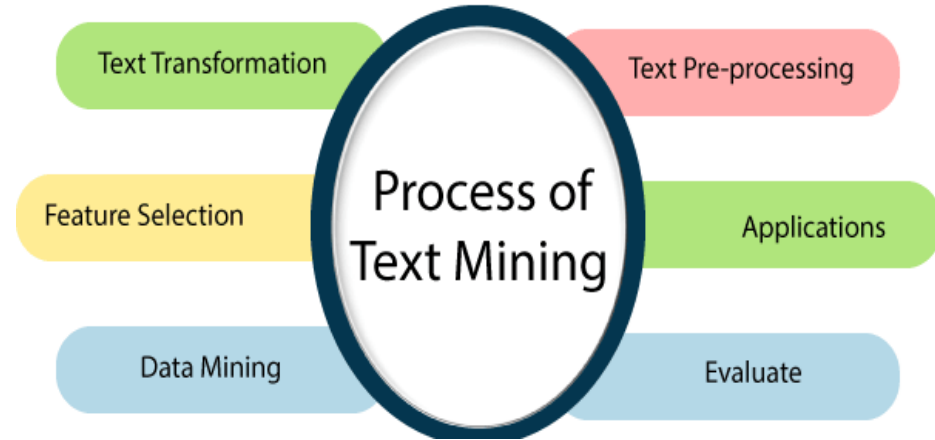
# **Data Warehousing and Data Mining**

## **Block-4 Unit-12 Part-2**

### **Text & Web Mining**

# Topics to be Covered

- Introduction
- Dimensionality Reduction
- Web Mining
- Types of Web Mining
- Mining Multimedia Data on the Web
- Automatic Classification of Web Documents
- Important Topics
- Summary



<https://www.javatpoint.com/text-data-mining>

# Introduction

- In this session we will focus on :-

The goal of text and web mining is to make sense of unstructured data and convert it into structured data that can be easily analyzed and used for various applications.

# Dimensionality Reduction

- The number of input features, variables, or columns present in a given dataset is known as dimensionality, and the process to reduce these features is called dimensionality reduction.
- Dimensionality reduction is the process of reducing the number of random variables or attributes under consideration. High-dimensionality data reduction, as part of a data pre-processing-step, is extremely important in many real-world applications.
- High dimensionality reduction has emerged as one of the significant tasks in data mining applications.



# Dimensionality Reduction

- **Curse of Dimensionality**

- Handling the high-dimensional data is very difficult in practice, commonly known as the curse of dimensionality.
- If the dimensionality of the input dataset increases, any machine learning algorithm and model becomes more complex.
- As the number of features increases, the number of samples also gets increased proportionally, and the chance of over fitting also increases.
- If the machine learning model is trained on high-dimensional data, it becomes over fitted and results in poor performance.

# Dimensionality Reduction

- **Benefits of applying dimensionality reduction technique**
  - By reducing the dimensions of the features, the space required to store the dataset also gets reduced.
  - Less Computation training time is required for reduced dimensions of features.
  - Reduced dimensions of features of the dataset help in visualizing the data quickly.
  - It removes the redundant features (if present) by taking care of multi-collinearity.

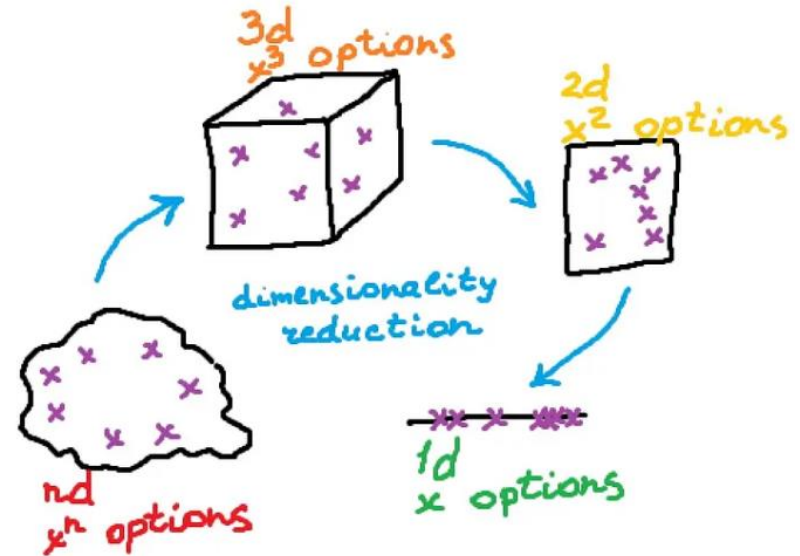
# Dimensionality Reduction

- **Techniques for Dimensionality Reduction**
- Dimensionality reduction is accomplished based on either feature selection or feature & extraction.
  - **Feature selection** It is based on omitting those features from the available measurements which do not contribute to class separability. In other words, redundant and irrelevant features are ignored.
  - **Feature extraction** The whole information content and maps the useful information content into a lower dimensional feature space.

# Dimensionality Reduction

## Feature Selection Techniques

- Variance Thresholds
- Correlation Thresholds
- Genetic Algorithms
- Stepwise Regression



# Dimensionality Reduction

- **Feature Extraction Techniques**
  - Linear Discriminant Analysis (LDA)
  - Principal Component Analysis (PCA)
  - t-distributed Stochastic Neighbor Embedding (t-SNE)
  - Auto encoders
    - Encoder
    - Decoder

# Web Mining

- **Web Mining** can be referred as discovering interesting and useful information from Web content and usage.
- Web mining is involves the mining of web data & extraction of information from websites uses data mining techniques. It is an application based on data mining techniques. Features :-
- Web search, e.g. Google, Yahoo, MSN, Ask, job ads .
- The web mining is not like relation, it has text content and linkage structure.
- Web mining can react in real-time with dynamic patterns generated on the web.

# Web Mining Tasks

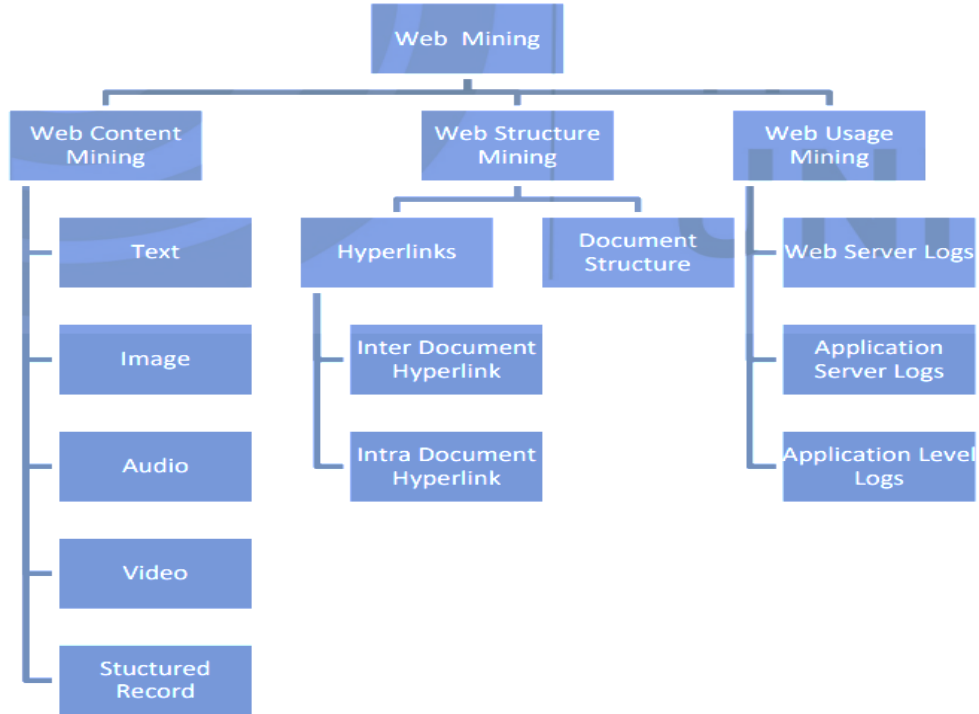
- **Web Mining performs various tasks such as:**
  - Generating patterns existing in some websites, like customer buying behavior or navigation of web sites.
  - The web mining helps to retrieve faster results of the queries or the search text posted on the search engines like Google, Yahoo etc.
  - The ability to classify web documents according to the search performed on the ecommerce websites helps to increase businesses and transactions.

# Web Mining Tasks

- **Applications of Web Mining**
  - Personalized customer experience in Business to Consumer (B2C)
  - Web Search
  - Web-wide tracking
  - Understanding Web Communities
  - Fraud detection
  - Improving Web site design and performance
  - Recommendations: e.g. Netflix, Amazon



# Types of Web Mining

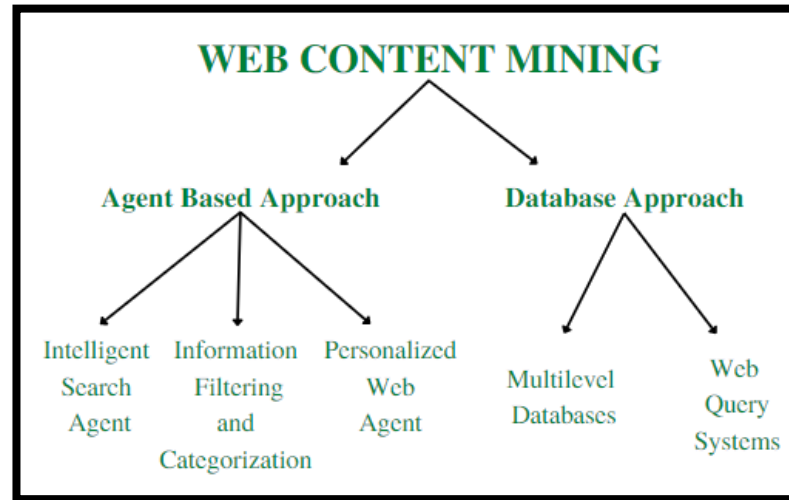


# Web Content Mining

- **Web content mining** is the process of extracting useful information from the contents of web documents. Content data is the collection of facts a web page is designed to contain. It may consist of text, images, audio, video, or structured records such as lists and tables.
- **Web Structure Mining** The structure of a typical web graph consists of web pages as nodes, and hyperlinks as edges connecting related pages. Web structure mining is the process of discovering structure information from the web. Web Structure Mining is further divided into two kinds
  - Hyperlinks
  - Document Structure

# Web Content Mining

- **Web Usage Mining** is the application of data mining techniques to discover interesting usage patterns from web usage data, in order to understand and better serve the needs of web-based applications.



<https://www.geeksforgeeks.org/what-is-web-content-mining/>

# Web Content Mining

Web usage mining itself can be classified further depending on the kind of usage data

Web Server  
Data

Application  
Server Data

Application  
Level Data

# Mining Multimedia Data on The Web

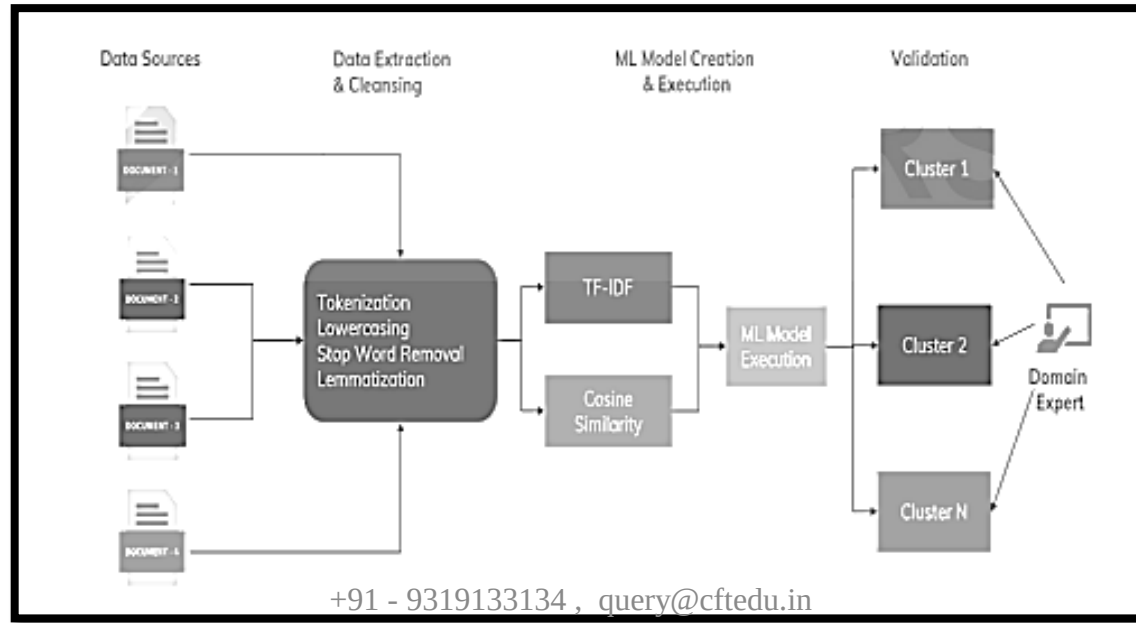
- The websites are flooded with the multimedia data like, video, audio, images, and graphs. This multimedia data has different characteristics. The videos, images, audio, and pictures have different methods of archiving and retrieving the information.
- **PageRank:** This measure is used to count the number of pages the webpage is connected to other websites. It gives the importance of the webpage.
- **HITS:** This measure is used to rate the webpage. It was developed by Jon Kleinberg.

# Mining Multimedia Data on The Web

- **Page Layout Analysis:** It extracts and maintains the page-to-block, block-to-page relationships from link structure of web pages.
- **Vision page segmentation (VIPS) algorithm:** It first extracts all the suitable blocks from the HTML Document Object Model (DOM) tree, and then it finds the separators between these blocks.

# Automatic Classification of Web Documents

- The categorization of web pages into the respective subjects or domains is called classification of web documents.



# Automatic Classification of Web Documents

- For example it has shown various categories like, books, electronics etc. let's say you are doing online shopping on the Amazon website and there are so many webpages so when you search for electronics the respective web page containing the information of electronics is displayed. This is the classification of products which is done on the textual and image contents.

## Categories



Books



Electronics



Home &  
Kitchen



Clothing



Movies



Health



Sports



# Automatic Classification of Web Documents

- The problem with the classification of web documents is that every time the model is to be constructed by applying some algorithms to classify the document is mammoth task.
- Automated document classification identifies the documents & groups the relevant documents without any external efforts.
- There are various tools available in the market like Rapid Miner, Azure, Machine Learning Studio, Amazon Sage maker, KNIME and Python. The trained model automatically reads the data from documents (PDF, DOC, PPT) and classifies the data according to the category of the document.

# Automatic Classification of Web Documents

- **Benefits of Automatic Document Classification System**
  - It is more efficient system of classification as produces improved accuracy of results and speed up the process of classification.
  - The system incurs in less operational costs easy data store and retrieval.
  - Easy data store and retrieval.
  - It organizes the files and documents in a better streamlined way.

# Important Topics

- Text Mining and its applications
- Text preprocessing
- Bow and TF-IDF For Creating Features from Text

**IMPORTANT**

<https://www.collegedekho.com/articles/neet-ug-most-important-topics/>

+91 - 9319133134 , [query@cftedu.in](mailto:query@cftedu.in)

# Summary

- **In this session we have learned about :-**
- Text mining which is, also referred to as text analysis, is the process of obtaining meaningful information from large collections of unstructured data.
- By automatically identifying patterns, topics, and relevant keywords, text mining uncovers relevant insights that can help you answer specific questions.
- Text mining makes it possible to detect trends and patterns in data that can help businesses support their decision-making processes.

*Thank  
You*